

Covariate Adjustment in Randomized Experiments: A Unified Framework for Decision and Practice

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Abstract

Should researchers adjust for covariates in randomized experiments, and if so, how? The literature offers three distinct prescriptions: do not adjust because randomization guarantees unbiasedness; adjust for outcome-prognostic covariates to improve precision; or adjust for covariates imbalanced between treatment arms. These competing prescriptions create confusion and uncertainty. We develop a unified framework for decision and practice. Given available information, we show that the optimal correction is what we call *ex-post bias*. The only relevant criterion for adjustment is prognosticity for ex-post bias; neither raw covariate imbalance nor outcome prognosticity is sufficient by itself. We also show that correcting imbalance and improving precision are two sides of the same decision problem. We develop two estimation approaches, one of which recovers familiar adjustment estimators and provides a new theoretical justification for them. Simulations compare alternative covariate-selection and adjustment strategies. Overall, our framework provides a unified foundation for covariate adjustment in randomized experiments.

Keywords: randomized experiments; covariate adjustment; covariate balance; prognostic scores; finite-population inference

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1 Introduction

Suppose that a randomized experiment has been completed. When analyzing the resulting data, should the researcher adjust for pretreatment covariates? If so, which covariates should be used, and how should the adjustment be carried out? These questions appear routine, but the usual answers rest on different ideas about what covariate adjustment is supposed to accomplish.

There exists at least three distinct ideas. One approach is to make no adjustment and report the unadjusted difference in means because properly implemented randomization makes this estimator unbiased (Meier, 1993). A second approach is to adjust when covariates are imbalanced, on the grounds that the two groups may not be comparable in the realized assignment (Johansson and Nordin, 2022). A third approach is to adjust for covariates that predict outcomes because prognostic adjustment can reduce residual variation and increase precision (Zhang et al., 2008). These distinct rationales lead to diverse practices. Researchers sometimes retain the unadjusted estimator and treat it as the gold-standard estimate. In other cases, they use a balance table to identify imbalanced covariates, select prognostic covariates using outcome models, and include the selected variables in a regression. We collect field experiments published in three leading political science journals from 2016 to 2025 and summarize their approaches to covariate adjustment in Table 1. As the table shows, all three rationales are commonly invoked in practice.

Table 1: Reported rationales in leading journals from 2016 to 2025

Journal	Papers	R1: No adjustment		R2: Imbalance	R3: Prognosticity
		Explicit rationale	Reports DIM	correction	/ precision
AJPS	41	3 (7.3%)	21 (51.2%)	4 (9.8%)	15 (36.6%)
APSR	38	4 (10.5%)	14 (36.8%)	7 (18.4%)	7 (18.4%)
JOP	31	3 (9.7%)	15 (48.4%)	5 (16.1%)	6 (19.4%)
Total	110	10 (9.1%)	50 (45.5%)	16 (14.5%)	28 (25.5%)

Notes: Entries are paper counts, with within-journal percentages in parentheses. R1 (explicit rationale) records an explicit appeal to randomization, unbiasedness, or the lack of a need for controls. R1 (reports DIM) records raw post-treatment mean/proportion comparisons or treatment-only OLS, whether or not an explanation is provided. R2 and R3 count directly expressed arguments. Categories overlap, and a paper is counted if the criterion is met in at least one reviewed study or analysis.

We review a collection of experimental studies published between 2016 and 2025 in three leading political science journals and summarize their stated rationales for covariate adjustment and their reporting of unadjusted estimates in Table 1. All three rationales are frequently invoked in practice. But are these rationales well founded, and which one should guide researchers' adjustment decisions? Existing debates often talk past one another. A clear, unified answer is therefore needed. This paper develops such a framework. We show that each argument contains important insights, but conventional accounts also miss the central roles of bias, imbalance, and prognosticity. The framework clarifies these concepts and provides practical guidance.

Our decision framework begins with a practical concern. Although the difference-in-means estimator $\hat{\tau}$ is ex-ante unbiased for the average treatment effect (ATE), researchers usually observe only a single experiment, and its realized estimate will generally differ from the true effect τ because the treatment and control groups are not perfectly balanced. Researchers often mistakenly

believe that a single randomized experiment balances all covariates and therefore returns the exact causal effect. In fact, a gap generally remains between the true effect and the estimate from a realized experiment. Can available information reveal this *ex-post* randomization error $e = \hat{\tau} - \tau$ and support a correction that moves the estimate closer to the true effect? We call the error *ex post* because it is considered after the randomization has been conducted, when researchers may have learned information $\mathcal{I}$ about the randomization through a balance check. For example, researchers may learn that a covariate mean is larger in the treatment group and therefore become concerned that the realized estimate deviates from the unobserved true effect.

We formalize and solve the decision problem. We show that the optimal correction is exactly the *ex-post bias* $\mathbb{E}_Z[e|\mathcal{I}]$. Estimating and subtracting this term can remove the *ex-post* bias. From an *ex-ante* perspective, the resulting estimator has lower variance than the unadjusted estimator. This framework thus unifies two objectives that are often treated separately. From an *ex-post* perspective, adjustment removes predictable error in the particular experiment that occurred. From an *ex-ante* perspective, the same adjustment reduces variation over the randomization distribution and increases precision. Therefore, concerns about covariate imbalance and precision are not competing justifications for covariate adjustment; rather, they are two manifestations of the same underlying decision problem. They are two interpretations of the same underlying prediction problem.

Our key result is that the relevant criterion for covariate adjustment is prognosticity for the randomization error, or equivalently, the *ex-post* bias. Neither raw covariate imbalance nor prognosticity for the observed outcome is, by itself, the fundamental criterion for adjustment. This result also clarifies the relevant notions of *prognosticity* and *imbalance*, showing that conventional interpretations are valid only in special cases. Traditionally, a covariate is considered prognostic if it predicts the observed outcome, the control potential outcome, or either treatment-specific potential outcome. For ATE estimation, however, the relevant question is more specific: does the covariate predict the *ex-post* randomization error? We provide examples in which a covariate is imbalanced and strongly predicts outcomes in both treatment arms, yet contains no information about the ATE estimation error. Conversely, a covariate that predicts the randomization error can be valuable for adjustment even if it is not a strong predictor of either potential outcome considered separately.

Similarly, we distinguish raw covariate imbalance from ATE-relevant imbalance. Raw covariate differences describe how treatment assignment is distributed across observed baseline characteristics. Such differences are valuable for transparency and for diagnosing potential failures in experimental implementation, such as those arising from attrition. However, raw covariate imbalance does not necessarily imply *ex-post* bias and therefore does not, by itself, justify covariate adjustment. We define ATE-relevant imbalance and show that it is this form of imbalance, rather than raw covariate imbalance, that is informative for adjustment.

Because the *ex-post* error is unobserved, we introduce two approaches to estimating it. Perhaps surprisingly, one of these approaches leads to familiar covariate-adjustment estimators. Therefore, conventional regression adjustment, generalized regression, augmented estimators, and flexible machine-learning procedures can all be used to construct the prognostic score (Lin, 2013; Robins et al., 1994). Our framework thus provides a new theoretical justification for these methods. What differs is that our framework highlights that the decision to adjust for a covariate depends only on its prognosticity for *ex-post* randomization error. Our second approach estimates this target more

directly and, in our finite-sample simulations, outperforms existing adjustment estimators. One reason is that the direct approach requires estimating only a single nuisance function, allowing it to use the available data more efficiently and potentially reducing estimation noise relative to arm-specific approaches that estimate separate nuisance functions for each treatment arm.

We evaluate the framework in a main simulation with $N = 400$ and $K = 60$. We compare a range of estimators across four settings: no useful information about the ATE estimation error; a sparse systematic component that depends directly on four of the sixty covariates; strong within-arm outcome relationships that cancel out for ATE adjustment; and unequal treatment allocation with heterogeneous treatment effects. We also vary sample size ($N \in \{160, 240, 400\}$) to examine finite-sample estimation performance. Overall, our direct estimator has the strongest performance profile in these simulations. The arm-specific AIPW estimator remains a credible alternative when separate outcome models are substantively desirable.

The supplementary study examines covariate selection policy. It compares five rules: control-outcome prediction (prognosticity), covariate imbalance, their intersection, their union, and selection under the proposed design-relevant weighted loss. The study confirms our theories that control-outcome prognosticity and/or covariate imbalance is not sufficient for prognosticity with respect to ATE estimation error; the design-relevant weighted criterion is preferable among the rules studied.

Our unified framework provides clear guidance on both *when* and *how* researchers should adjust. Researchers concerned with ex-post bias should adjust whenever covariates are prognostic for that bias using adjustment estimators. In practice, researchers should collect a rich set of pretreatment covariates. Because ex-post bias is inherently unobserved, a useful minimum guideline is to collect covariates that are plausibly prognostic for the potential outcomes. Our proposed estimators, combined with machine-learning methods, can use these covariates to adaptively approximate the relevant component of ex-post bias and produce an adjusted ATE estimate. Researchers should not decide whether to adjust based on tests of whether individual raw covariates happen to be balanced in the realized assignment or solely on whether those covariates predict control-group outcomes. At the same time, we do not recommend abandoning conventional balance tables. Raw covariate balance remains useful for describing the experiment, assessing implementation of the assignment mechanism, and identifying unusual features of the realized sample. A conventional balance table should therefore be retained but supplemented by an outcome- and estimand-specific prognostic balance report.

The remainder of the paper proceeds as follows. We first review the competing arguments surrounding covariate balance and adjustment. We then develop the decision framework, derive feasible prognostic-adjustment procedures, and discuss their implications for balance reporting and existing methods. Finally, we evaluate the framework through simulations and conclude.

2 Debates over Covariate Balance and Adjustment

Should researchers adjust for pretreatment covariates in randomized experiments? If so, should covariates be selected because they are imbalanced in the realized assignment, because they predict the outcome, or for some combination of these reasons? Despite the widespread use of regression adjustment in experimental research, there is no single consensus answer supported by a common

rationale.

The apparent disagreement in the literature partly reflects the fact that several distinct questions are often discussed under the common heading of “covariate adjustment.” Randomization addresses the marginal validity of an estimator over all possible assignments. Balance diagnostics describe features of the assignment that was actually realized. Prognostic adjustment seeks to improve precision by explaining outcome variation. These objectives are related, but they are not interchangeable.

This section reviews these three rationales and then explains how the decision framework developed below places them within a common structure while highlighting what each perspective overlooks.

Ex-Ante Unbiasedness. One influential view begins from a simple principle: in a properly randomized experiment, the difference-in-means estimator is unbiased for the average treatment effect in expectation over all possible randomizations. Covariate adjustment is therefore not required for identification. This feature makes the unadjusted estimator transparent and provides a natural benchmark against which adjusted estimators should be evaluated.

Concerns about conventional regression adjustment reinforce this position. [Freedman \(2008b,a\)](#) show that ordinary least squares can have finite-sample bias under the randomization model, that the usual model-based variance formulas need not be justified, and that adjustment need not always improve precision. These results caution against treating a randomized experiment as if it were generated by a correctly specified linear regression model. Subsequent work, however, substantially qualifies these concerns. With treatment–covariate interactions and heteroskedasticity-robust standard errors, regression adjustment is asymptotically valid and no less efficient than the unadjusted estimator under standard regularity conditions ([Lin, 2013](#)). Exact or higher-order bias corrections can further address the finite-sample bias emphasized by Freedman ([Wu and Gagnon-Bartsch, 2018](#); [Chang et al., 2024](#)).

Covariate Imbalance. A second perspective agrees that randomization eliminates systematic bias over the assignment distribution. However, because an experiment is conducted only once, a particular random assignment may place units with systematically more favorable prognoses in one treatment arm. Conditional on the resulting signed covariate imbalance, the difference-in-means estimator can therefore have a nonzero conditional mean error even though its unconditional randomization bias is zero.

This concern has a long history. Early discussions of clinical-trial analysis describe covariate adjustment as a way to account for baseline variables on which randomization failed to achieve close balance ([Beach and Meier, 1989](#); [Pocock et al., 2002](#)). In practice, balance reporting is therefore common in experimental research and is encouraged by reporting standards in political science and other fields ([Gerber et al., 2014](#)). Randomization-based omnibus tests have also been developed to summarize whether the observed treatment–control differences are unusual under the specified assignment mechanism ([Hansen and Bowers, 2008](#)).

A further step is to choose covariates for adjustment on the basis of a balance test. This practice is much more controversial. [Senn \(1994\)](#) argues that significance tests of baseline equality are uninformative and potentially misleading in randomized trials. Moreover, adjustment after balance test can have undesirable efficiency and inferential properties ([Mutz et al., 2019](#); [Zhao and Ding,](#)

2024).

Prognostic Adjustment and Efficiency. A third and broader school justifies adjustment on efficiency grounds. The key principle is prognosticity: baseline information is useful when it predicts components of the potential outcomes that contribute to sampling or randomization error. Prespecified adjustment for such covariates can reduce residual outcome variation and increase precision.

This principle underlies classical analysis of covariance and modern semiparametric augmentation methods (Tsiatis et al., 2008; Zhang et al., 2008). In the design-based literature, fully interacted regression adjustment is asymptotically no less efficient than the difference in means (Lin, 2013; Negi and Wooldridge, 2021). Related work shows that substantial gains are possible when baseline variables are strongly predictive of outcomes (Colantuoni and Rosenblum, 2015). High-dimensional methods extend this logic by using regularization, cross-estimation, or machine learning to estimate prognostic components while controlling the cost of overfitting (Bloniarz et al., 2016; Wager et al., 2016; Wu and Gagnon-Bartsch, 2018). Current FDA guidance likewise focuses on prespecified prognostic baseline covariates as a means of increasing precision and power.¹

Summary. The debate remains unresolved. Each school contains correct but incomplete insights. The ex-ante unbiasedness school correctly observes that randomization eliminates systematic confounding in expectation, but it overlooks efficiency and the fact that each realized experiment has its own error. The imbalance school correctly recognizes that covariate imbalance can matter in a single realized experiment. In practice, however, balance-test-and-adjust procedures can generate additional bias and overlook potential efficiency gains. The efficiency school emphasizes prognosticity and rests on a strong theoretical foundation, but it gives less attention to bias associated with covariate imbalance, which is a principal concern motivating balance tables in experimental research.

These schools of thought are not mutually exclusive; rather, each addresses a different part of the adjustment problem. The ex-ante perspective evaluates an estimator over repeated assignments, whereas empirical analysis confronts one realized randomization. Budget constraints, ethical limitations, and the irreversibility of many treatments make the realized experiment, rather than the hypothetical average over all possible assignments, the relevant practical object. The framework developed in the next section connects these practical concerns and perspectives by formalizing when and how a realized estimate should be corrected. We show that the latter two perspectives are two sides of the same coin. Our central guideline is that the relevant criterion for covariate adjustment is prognosticity for ex-post bias. Neither raw covariate imbalance nor conventional outcome prognosticity is, by itself, the fundamental criterion for adjustment.

3 A Decision Framework with Ex-Post Bias

Consider a finite population of N units. Unit i has fixed potential outcomes $Y_i(1)$ and $Y_i(0)$ under treatment and control, respectively, and a K -dimensional vector of pretreatment covariates $X_i = (X_i^1, \dots, X_i^K)^\top \in \mathcal{X}$. Let $Z_i \in \{0, 1\}$ denote treatment assignment. We assume

¹<https://www.fda.gov/regulatory-information/search-fda-guidance-documents/adjusting-covariates-randomized-clinical-trials-drugs-and-biological-products>

complete randomization with exactly N_1 treated units and $N_0 = N - N_1$ control units, so Z is drawn uniformly from $\mathcal{Z}_{N_1} = \{z \in \{0, 1\}^N : \sum_i z_i = N_1\}$. The observed outcome is $Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$, and the finite-population ATE is $\tau = N^{-1} \sum_{i=1}^N \{Y_i(1) - Y_i(0)\}$.

The unadjusted difference-in-means estimator is $\hat{\tau}_{\text{DM}} = \hat{\tau} = \frac{1}{N_1} \sum_{i=1}^N [Z_i Y_i] - \frac{1}{N_0} \sum_{i=1}^N [(1 - Z_i) Y_i]$. Complete randomization makes this estimator unbiased over all possible assignments. Its value under a particular assignment, however, generally differs from the finite-population ATE. As [Deaton and Cartwright \(2018\)](#) emphasize, errors may cancel across a hypothetical sequence of randomizations even though the estimate from the realized experiment is far from the target. This observation motivates a distinction between the estimator’s ex-ante bias and its realized randomization error.

We first formalize the assignment-specific discrepancy between an estimator and the true ATE.

Definition 1 (Realized randomization error). For a realized assignment $Z = z$, the realized error of an estimator $\hat{\tau}$ is

$$e(z) = \hat{\tau}(z) - \tau. \quad (1)$$

For the unadjusted estimator, we write $e_0(Z) = \hat{\tau}_{\text{DM}}(Z) - \tau$. Each possible treatment assignment induces an estimate $\hat{\tau}$ and thus an error $e(z)$. For example, as shown in Table 2, the first randomization generates an error of 7, whereas the second generates an error of -8 . In practice, researchers cannot observe this oracle randomization table or know the realized error. However, because of randomization, the expected error over all possible assignments is zero ex ante: $\mathbb{E}_Z\{e(Z)\} = 0$. This does not imply that $e(z) = 0$ for every $z \in \mathcal{Z}_{N_1}$.

Randomization	Imbalanced covariates	Error ($e(z)$)
1	X^2, X^5, X^{10}	7
2	X^1, X^4, X^5, X^7	-8
3	X^2, X^{10}	-4
$\vdots$		
$\binom{N}{N_1}$	$X^1, X^2, X^6, X^9, X^{10}$	5

Table 2: An artificial randomization table known to the oracle

Because the realized randomization error is unknown, researchers may use observed covariates and balance information to learn about it. For motivation, suppose researchers learn that X_2 is imbalanced in the realized assignment. This observation rules out some randomizations, including the second row of Table 2, but remains compatible with many others, such as rows 1, 3, and the final row. Let $\mathcal{I}$ denote the information generated by this or any other prespecified post-assignment report. We define the corresponding ex-post bias as follows.

Definition 2 (Ex-post bias). The ex-post bias given the information set $\mathcal{I}$ is

$$B_{\mathcal{I}} := \mathbb{E}_Z[\hat{\tau} - \tau \mid \mathcal{I}].$$

Thus, ex-post bias averages the randomization errors of assignments that are indistinguishable given $\mathcal{I}$.

Our decision framework begins with a practical concern. After observing $\mathcal{I}$, the researcher must decide whether to correct the unadjusted estimator and, if so, by how much. Consider the estimator

$$\hat{\tau}_a = \hat{\tau}_{\text{DM}} - a(\mathcal{I}), \quad (2)$$

where $a(\mathcal{I})$ is any square-integrable $\mathcal{I}$ -measurable correction.

We adopt mean-squared error as the decision criterion. Conditional MSE evaluates a correction among assignments consistent with the observed information, whereas unconditional MSE evaluates it before treatment assignment over the full randomization distribution. The following theorem shows that both criteria identify the same optimal correction and quantifies the loss from any alternative action.

Proposition 1 (Decision theorem). *For every $\mathcal{I}$ -measurable action $a(\mathcal{I})$,*

$$\text{MSE}_Z(\hat{\tau}_{\text{DM}}|\mathcal{I}) - \text{MSE}_Z(\hat{\tau}_a|\mathcal{I}) = B_{\mathcal{I}}^2 - [B_{\mathcal{I}} - a(\mathcal{I})]^2 \quad (3)$$

$$\text{MSE}_Z(\hat{\tau}_{\text{DM}}) - \text{MSE}_Z(\hat{\tau}_a) = \text{Var}_Z(B_{\mathcal{I}}) - \mathbb{E}[B_{\mathcal{I}} - a(\mathcal{I})]^2 \quad (4)$$

Consequently, the unique MSE minimizer is

$$a^*(\mathcal{I}) = B_{\mathcal{I}}. \quad (5)$$

Setting $a = a^*$ in Equation (4) gives the oracle MSE gain directly:

$$\text{MSE}_Z(\hat{\tau}_{\text{DM}}) - \text{MSE}_Z(\hat{\tau}_{a^*}) = \text{Var}_Z(B_{\mathcal{I}}) \geq 0. \quad (6)$$

The proposition separates the information from the quality of a correction based on it. The first feature of Equations (3) and (4) is the *value of information*, captured by $B_{\mathcal{I}}$ for a given information set and, across treatment assignments, by $\text{Var}_Z(B_{\mathcal{I}})$. The random variable $B_{\mathcal{I}}$ is the component of the unadjusted randomization error that can be predicted from $\mathcal{I}$. Because $\mathbb{E}_Z(B_{\mathcal{I}}) = 0$, its variance, $\text{Var}_Z(B_{\mathcal{I}}) = \mathbb{E}_Z(B_{\mathcal{I}}^2)$, is the average squared magnitude of that predictable component across assignments. Equivalently, as shown in equation (6), it is the portion of the unadjusted randomization MSE that an oracle could remove by using $\mathcal{I}$ perfectly. Thus, removing conditional ex-post bias and improving unconditional randomization precision are two interpretations of the same projection.

The second feature is the *quality of the chosen correction*. The discrepancy $B_{\mathcal{I}} - a(\mathcal{I})$ measures how far the proposed action is from the oracle correction. Equation 4 therefore says that the net MSE gain equals the informational value of the report minus the approximation error of the action. A feasible correction improves on the unadjusted estimator if and only if $\mathbb{E}[B_{\mathcal{I}} - a(\mathcal{I})]^2 < \text{Var}_Z(B_{\mathcal{I}})$. Thus, the mere existence of information does not guarantee that an arbitrary adjustment based on it will help. A poorly chosen correction can exhaust the report's informational value and can even increase MSE. At the optimum $a = a^*$, the approximation cost vanishes, so the correction removes the conditional ex-post bias and achieves the maximum unconditional MSE gain.

Two implications follow. If $\text{Var}_Z(B_{\mathcal{I}}) = 0$, then $\mathcal{I}$ contains no information about the *conditional mean* of the randomization error, and every nonzero additive correction based only on that information weakly increases MSE. Conversely, richer information can only improve the oracle decision. The next proposition formalizes this monotonicity.

Proposition 2 (Value of finer information). *Let $\mathcal{I}_1 \subseteq \mathcal{I}_2$ and for $j \in \{1, 2\}$ define $B_j = \mathbb{E}_Z(e_0 \mid \mathcal{I}_j)$, and $\text{Risk}^*(\mathcal{I}_j) = \mathbb{E}_Z\{(e_0 - B_j)^2\}$. Then $B_1 = \mathbb{E}_Z(B_2 \mid \mathcal{I}_1)$, and*

$$\text{Risk}^*(\mathcal{I}_1) - \text{Risk}^*(\mathcal{I}_2) = \mathbb{E}_Z\{(B_2 - B_1)^2\} = \text{Var}_Z(B_2) - \text{Var}_Z(B_1) \geq 0. \quad (7)$$

Proposition 2 shows that, at the oracle level, retaining finer information cannot worsen prediction of the bias. We now formalize the information and introduce three layers of information. Begin with the finest information generated by the observed covariate profiles. Let $x^{(1)}, \dots, x^{(J)}$ be the distinct observed covariate profiles in the finite population, and define $\mathcal{C}_j = \{i : X_i = x^{(j)}\}$. For a realized assignment, let $N_{1j}(Z) = \sum_{i \in \mathcal{C}_j} Z_i$. The vector $H_X(Z) = \{N_{11}(Z), \dots, N_{1J}(Z)\}$ records the complete treatment allocation across the observed covariate profiles. We define $\mathcal{F}_X = \sigma\{H_X(Z)\}$ as the finest post-assignment information generated by the observed covariate profiles. The sigma-field $\mathcal{F}_X$ records the complete allocation of treatment across the observed covariate profiles, but not which unit labels within a profile are treated.

The second layer incorporates pretreatment characteristics that researchers do not observe. Let U_i denote characteristics that are unobserved or unrecorded, and let $V_i = (X_i, U_i)$ denote the full covariate profile. Let $\mathcal{F}_{\text{full}}$ be defined analogously from the treatment counts within the distinct full profiles V_i . Because every observed X -cell is a union of finer (X, U) -cells², $\mathcal{F}_X \subseteq \mathcal{F}_{\text{full}}$.

The third and generally coarsest layer is the particular balance report observed by the researcher. We use $T(Z, X)$ to denote any post-assignment statistic constructed from the realized assignment and pretreatment observables and define $\mathcal{I} = \sigma\{T(Z, X)\}$. Examples of T include the full vector of signed covariate mean differences, $T = \{\Delta_Z(X^1), \dots, \Delta_Z(X^K)\}$, the corresponding vector of absolute mean differences, an omnibus Mahalanobis balance statistic, a vector of two-sided balance-test p -values, or indicators of whether prespecified tests reject.

The information sets satisfy the following hierarchy: $\mathcal{I} \subseteq \mathcal{F}_X \subseteq \mathcal{F}_{\text{full}}$. This hierarchy clarifies the source of the remaining uncertainty. Conditional on $\mathcal{F}_X$, the numbers treated within each observed X -profile are fixed, but the allocation of treatment across the unobserved U -subprofiles inside an X -profile remains random. Equivalently, two assignments can belong to the same atom of $\mathcal{F}_X$ while belonging to different atoms of $\mathcal{F}_{\text{full}}$.

According to Proposition 2, we therefore focus on the observed-covariate information set $\mathcal{F}_X$ in what follows. That is, we use the observed covariates directly as the information available for constructing the adjustment, rather than prespecifying particular functions of them. This approach is both natural and flexible, as the adjustment procedure can learn relevant functions of the observed covariates from the data.

3.1 ATE-Relevant Imbalance and Prognosticity

The decision framework shows that the central task is to estimate ex-post bias. This subsection proceeds in two steps. First, we derive an exact score representation of the unadjusted randomization error and ex-post bias. Second, we show that approximating ex-post bias is equivalent to predicting this score up to an additive constant. Together, these results transform the decision problem above into a covariate-prediction problem.

²In particular, if $\mathcal{L}(j) = \{\ell : v^{(\ell)} = (x^{(j)}, u) \text{ for some } u\}$, then $N_{1j}^X(Z) = \sum_{\ell \in \mathcal{L}(j)} N_{1\ell}^V(Z)$. Hence $H_X(Z)$ is measurable with respect to $H_V(Z)$.

To state the score representation, we first introduce notation for treatment–control imbalance. For any fixed scalar finite-population variable $A = (A_1, \dots, A_N)$, define

$$\bar{A} = \frac{1}{N} \sum_{i=1}^N A_i, \quad \bar{A}_1 = \frac{1}{N_1} \sum_{i=1}^N Z_i A_i, \quad \bar{A}_0 = \frac{1}{N_0} \sum_{i=1}^N (1 - Z_i) A_i. \quad (8)$$

Its signed treatment–control imbalance is $\Delta_Z(A) = \bar{A}_1 - \bar{A}_0$.

Definition 3 (Design-weighted randomization-error). For treatment fraction $p = N_1/N$, define

$$M_i = (1 - p)Y_i(1) + pY_i(0). \quad (9)$$

The score M_i is an oracle object because the two potential outcomes are not jointly observed. Under equal allocation, $M_i = \frac{Y_i(1) + Y_i(0)}{2}$, so M_i is simply the midpoint of the two potential outcomes. Under unequal allocation, the midpoint is shifted toward the potential outcome observed in the smaller arm, whose sample composition has greater leverage on the difference-in-means estimator. The following proposition establishes that the realized treatment–control imbalance of M_i exactly equals the error of the unadjusted estimator.

Proposition 3 (Exact randomization-error identity). *Under complete randomization,*

$$\hat{\tau}_{\text{DM}} - \tau = \Delta_Z(M) = (1 - p)\Delta_Z\{Y(1)\} + p\Delta_Z\{Y(0)\}. \quad (10)$$

Equivalently, $\Delta_z(M) = \{\bar{Y}_1(1) - \bar{Y}(1)\} + \{\bar{Y}(0) - \bar{Y}_0(0)\}$. The first term is the error from using the realized treatment group to represent the population under treatment. The second is the error from using the realized control group to represent the population under control. $\Delta_z(M)$ asks: Did the realized treatment group contain units with systematically higher or lower ATE-relevant outcome levels than the realized control group? Accordingly, $\Delta_z(M) > 0$ means that the realized difference-in-means estimate overstates the ATE, while $\Delta_z(M) < 0$ means that it understates the ATE.

By Proposition 3, the oracle correction based on observed covariates is $B_X = \mathbb{E}_Z[\Delta_Z(M)|\mathcal{F}_X]$. We seek to approximate this complex quantity using the observed covariates X . We show that it is sufficient to approximate M up to a constant with function $g : \mathcal{X} \rightarrow \mathbb{R}$ constructed from X . The exact identity $e_0 = \Delta_Z(M)$ then suggests the plug-in correction $a_g = \Delta_Z\{g(X)\}$. The resulting estimator is

$$\hat{\tau}(g) = \hat{\tau}_{\text{DM}} - \Delta_Z\{g(X)\}$$

We now show that using $g(X)$ to approximate M up to a constant is equivalent to selecting $\Delta g(X)$ to approximate the optimal correction B_X . To state the equivalence formally, for two finite-population variables A and B , write $S_{AB} = \frac{1}{N-1} \sum_{i=1}^N (A_i - \bar{A})(B_i - \bar{B})$, $S_A^2 = S_{AA}$. For vector-valued X_i , S_{XX} denotes its finite-population covariance matrix and S_{XA} the vector of covariances between X and A .

Proposition 4. *Let $\mathcal{G}$ be a prespecified class of scores $g : \mathcal{X} \rightarrow \mathbb{R}$ that are held fixed with respect to the treatment assignment.*

$$\begin{aligned}
\arg \min_{g \in \mathcal{G}} \min_{c \in \mathbb{R}} \frac{1}{N} \sum_{i=1}^N \{M_i - c - g(X_i)\}^2 &= \arg \min_{g \in \mathcal{G}} \mathbb{E}_Z [\{\Delta_Z(M) - \Delta_Z\{g(X)\}\}^2] \\
&= \arg \min_{g \in \mathcal{G}} \mathbb{E}_Z [\{B_X - \Delta_Z\{g(X)\}\}^2] \\
&= \arg \min_{g \in \mathcal{G}} \text{MSE}_Z\{\hat{\tau}(g)\} \\
&= \arg \min_{g \in \mathcal{G}} S_{M-g}^2
\end{aligned} \tag{11}$$

The first and second equality show that choosing $g(X)$ to approximate M up to a constant is equivalent to choosing $\Delta_Z\{g(X)\}$ to approximate $\Delta_Z(M)$, which in turn minimizes the approximation error for the ex-post bias $\mathcal{B}_X$. Note that $\min_{c \in \mathbb{R}} \frac{1}{N} \sum_{i=1}^N \{M_i - c - g(X_i)\}^2 = \frac{1}{N} \sum_{i=1}^N [(M_i - \bar{M}) - (g(X_i) - \bar{g})]^2$. Thus, what matters is how well the centered values of $g(X)$ predict the centered values of M . We do not require $g(X)$ to fit the level of M perfectly: the two may differ by an arbitrary additive constant because only $\Delta_Z\{g(X)\}$ enters the correction.

The third equality further shows that the choice of $g(X)$ that best approximates the ex-post bias is also the choice that yields the optimal adjustment estimator. The fourth equality then brings the argument full circle: finding the best adjustment reduces to finding the function $g(X)$ that approximate M up to an additive constant. This yields our main take-home message: the relevant criterion for covariate adjustment is prognosticity for M . In particular, neither covariate imbalance nor prognosticity for the observed outcome is, by itself, the fundamental criterion for adjustment.

3.1.1 Covariate Imbalance and Outcome Prognosticity

Proposition 4 identifies the score that adjustment should target. It also reveals why raw covariate imbalance and generic outcome prognosticity are incomplete guides when considered separately. We now make these distinctions explicit.

Two common rationales for covariate adjustment focus on different properties of a pretreatment covariate. The balance-based rationale asks whether the covariate is distributed differently between the treatment and control groups in the realized assignment. The prognostic rationale asks whether the covariate predicts the outcome. Our framework shows that neither property, considered on its own, identifies whether the covariate is useful for estimating the ATE. A stronger point is that the two properties are insufficient even when they occur together.

The term *prognostic* ordinarily refers to pretreatment information that predicts an outcome. In the causal-inference literature, prognostic scores summarize the association between baseline covariates and potential responses (Hansen, 2008). In the randomized-trial literature, the standard precision argument likewise emphasizes adjustment for baseline covariates that are associated with the outcome (Tsiatis et al., 2008; Lin, 2013). The clinical biomarker literature further distinguishes a *prognostic* factor, which predicts the outcome level, from a *predictive* factor, which predicts differential treatment benefit and is therefore an effect modifier (Ballman, 2015).

These conventional uses are informative, but they do not fully answer the question studied here. For estimation of the ATE, it is not enough to ask whether X predicts $Y(1)$ or $Y(0)$ separately.

The relevant question is whether X predicts the particular combination

$$M_i = (1 - p)Y_i(1) + pY_i(0)$$

that determines the randomization error of the difference-in-means estimator.

The conventional notion of imbalance requires a similar refinement. A balance table typically reports treatment–control differences in individual covariates, $\Delta_z(X^1), \dots, \Delta_z(X^K)$, possibly accompanied by standardized differences or balance-test p -values. These quantities describe the realized assignment and can be useful for transparency and for diagnosing possible problems in experimental implementation. They do not, however, directly measure error in the estimated ATE. By Proposition 3, the imbalance that exactly determines the realized randomization error is $\Delta_z(M)$, not $\Delta_z(X^j)$ for an arbitrary raw covariate.

The following example illustrates both distinctions simultaneously. It echoes Corollary 1.1, Remark (i), of Lin (2013), but generalizes the insight to a broader setting and incorporates covariate imbalance into the analysis.

Example 1 (Offsetting Prognostic Imbalances under Unequal Allocation). *Consider an experiment with four units, one of which is assigned to treatment. Thus, $p = \frac{1}{4}$. Let $X_i = (X_i^1, X_i^2)^\top$ contain two pretreatment covariates, and suppose that the potential outcomes satisfy*

$$Y_i(1) = 24 + X_i^1 + 5X_i^2, \quad Y_i(0) = 16 + 5X_i^1 + X_i^2. \quad (12)$$

Both covariates therefore positively predict both potential outcomes. Consider the following finite-population science table:

i	1	2	3	4
X_i^1	2	−2	0	2
X_i^2	0	0	1	2
$Y_i(1)$	26	22	29	36
$Y_i(0)$	26	6	17	28

Suppose that unit 1 is assigned to treatment and units 2, 3, 4 are assigned to control. The two covariates are both imbalanced:

$$\Delta_z(X^1) = 2 - \frac{-2 + 0 + 2}{3} = 2, \quad (13)$$

$$\Delta_z(X^2) = 0 - \frac{0 + 1 + 2}{3} = -1. \quad (14)$$

Thus, a conventional balance table would report a positive imbalance in X^1 and a negative imbalance in X^2 .

The ATE-relevant randomization-error score, however, is

$$\begin{aligned} M_i &= \frac{3}{4}Y_i(1) + \frac{1}{4}Y_i(0) \\ &= \frac{3}{4}(24 + X_i^1 + 5X_i^2) + \frac{1}{4}(16 + 5X_i^1 + X_i^2) \\ &= 22 + 2X_i^1 + 4X_i^2. \end{aligned} \quad (15)$$

Consequently,

$$\begin{aligned}\Delta_z(M) &= 2\Delta_z(X^1) + 4\Delta_z(X^2) \\ &= 2(2) + 4(-1) \\ &= 0.\end{aligned}\tag{16}$$

The two raw covariate imbalances therefore offset after they are translated into their ATE-relevant prognostic contributions.

Indeed, the finite-population ATE is

$$\tau = \frac{1}{4} [(26 - 26) + (22 - 6) + (29 - 17) + (36 - 28)] = 9,$$

while the realized difference-in-means estimator is

$$\hat{\tau}_{\text{DM}} = 26 - \frac{6 + 17 + 28}{3} = 26 - 17 = 9.$$

Hence, $\hat{\tau}_{\text{DM}} - \tau = \Delta_z(M) = 0$, even though both covariates are prognostic for both potential outcomes and both are imbalanced in the realized assignment.

Example 1 establishes that raw covariate imbalance does not necessarily imply ATE-relevant imbalance:

$$\Delta_z(X) \neq 0 \not\Rightarrow \Delta_z(M) \neq 0.\tag{17}$$

It also establishes the stronger joint statement

$$[\Delta_z(X) \neq 0 \text{ and } X \text{ predicts } Y(1) \text{ and } Y(0)] \not\Rightarrow \Delta_z(M) \neq 0.\tag{18}$$

Crucially, the failure in this example is not that X lacks prognostic value for M . In fact, M is perfectly predicted by $g(X) = 22 + 2X^1 + 4X^2$. Rather, conventional measures of covariate imbalance do not aggregate covariate differences according to their ATE-relevant prognostic weights. Here, the positive contribution from the imbalance in X^1 is exactly offset by the negative contribution from the imbalance in X^2 . Thus, even a collection of genuinely prognostic and individually imbalanced covariates need not generate any realized error in the ATE estimate.

The relationship between prognosticity and realized imbalance becomes especially transparent for a linear score, $g(X_i) = \gamma^\top X_i$, the relevant feasible imbalance is

$$\Delta_z\{g(X)\} = \gamma^\top \Delta_z(X) = \sum_{j=1}^K \gamma_j \Delta_z(X^j).\tag{19}$$

Equation (19) makes clear why raw covariate imbalance is not sufficient. Each covariate difference matters only through its contribution to predicting M . A large imbalance in a covariate with no M -prognostic value receives zero weight. A smaller imbalance in a highly M -prognostic covariate can receive substantial weight. Imbalances in different covariates may also reinforce or offset one another depending on their signs and their prognostic coefficients.

Accordingly, the oracle score-selection problem is $g^* \in \arg \min_{g \in \mathcal{G}} S_{M-g}^2$, where the goal is not

necessarily to identify every variable that predicts either potential outcome, but rather to construct a score whose centered values predict the centered values of M .

The conceptual distinction can therefore be summarized as follows:

Correct prognostic question:	Does $g(X)$ predict M ?
Correct imbalance question:	Is $g(X)$ imbalanced in the realized assignment?

At the oracle level, the relevant imbalance is $\Delta_z(M)$. In practice, because M is unobserved, researchers estimate a score $\hat{g}(X) \approx M$ and report $\Delta_z\{\hat{g}(X)\}$. We return to this distinction in Section 6, where we develop an ATE-relevant prognostic balance report and explain how the score's predictive value and its realized imbalance should be reported separately.

4 Feasible Prognostic Adjustment

The preceding section identifies the oracle adjustment problem. Under complete randomization, $\hat{\tau}_{\text{DM}} - \tau = \Delta_Z(M)$, where $M_i = (1 - p)Y_i(1) + pY_i(0)$, and a pretreatment score $g(X)$ produces the adjusted estimator $\hat{\tau}(g) = \hat{\tau}_{\text{DM}} - \Delta_Z\{g(X)\}$. The difficulty is that M_i is not observed because $Y_i(1)$ and $Y_i(0)$ are never jointly observed.

There are two natural implementation routes. Researchers may learn the single score g directly from observed outcomes, or they may estimate the two arm-specific outcome functions and combine them using the complementary design weights. Both routes enter the treatment-effect estimator through the same score imbalance. In particular, the arm-specific route yields exactly the familiar generalized-regression or augmented inverse-probability-weighted estimator.

4.1 Direct Learning of the Single Score

To learn M without observing both potential outcomes, consider the design-weighted finite-population loss

$$\mathcal{L}_p(g) = (1 - p) \frac{1}{N} \sum_{i=1}^N \{Y_i(1) - g(X_i)\}^2 + p \frac{1}{N} \sum_{i=1}^N \{Y_i(0) - g(X_i)\}^2. \quad (20)$$

Its observable analogue is

$$\hat{\mathcal{L}}_p(g) = (1 - p) \frac{1}{N_1} \sum_{i=1}^N Z_i \{Y_i - g(X_i)\}^2 + p \frac{1}{N_0} \sum_{i=1}^N (1 - Z_i) \{Y_i - g(X_i)\}^2. \quad (21)$$

The complementary weights are chosen to match the potential-outcome weights in $M = qY(1) + pY(0)$. Under equal allocation, the observed criterion reduces to ordinary pooled squared-error loss:

$$\hat{\mathcal{L}}_{1/2}(g) = \frac{1}{N} \sum_{i=1}^N \{Y_i - g(X_i)\}^2.$$

Under unequal allocation, the complementary weights are essential. The criterion can be written as

$$\hat{\mathcal{L}}_p(g) = \frac{1}{N} \sum_{i=1}^N \omega_i \{Y_i - g(X_i)\}^2,$$

where

$$\omega_i = \begin{cases} q/p, & Z_i = 1, \\ p/q, & Z_i = 0. \end{cases}$$

Thus, the potential-outcome surface observed in the smaller treatment arm receives greater weight because its composition contributes more to the randomization error of the difference-in-means estimator.

The following proposition shows that the observable criterion targets precisely the prediction problem identified by the oracle MSE analysis.

Proposition 5 (Observed-outcome learning targets the oracle score). *Let $\mathcal{G}$ be a prespecified class of functions, and let $\mathcal{L}_p(g)$ be defined as in (20). For every $g \in \mathcal{G}$, $\arg \min_{g \in \mathcal{G}} \mathcal{L}_p(g) = \arg \min_{g \in \mathcal{G}} \frac{1}{N} \sum_{i=1}^N \{M_i - g(X_i)\}^2$.*

The proposition 5 identifies the oracle target. Define $g_{\mathcal{G}}^* \in \arg \min_{g \in \mathcal{G}} \mathcal{L}_p(g)$. Proposition 5 shows that $g_{\mathcal{G}}^*$ is the best predictor of M within $\mathcal{G}$. When $\mathcal{G}$ is closed under adding constants, it also induces the same treatment–control imbalance as the MSE-optimal centered predictor characterized in Proposition 4.

The final statement of Proposition 5 is pointwise in g . It says that the observed loss is unbiased for the oracle loss when a candidate function is held fixed over the randomization distribution. In practice, researchers estimate the score. In a low-dimensional class, a natural estimator is the empirical risk minimizer $\hat{g} \in \arg \min_{g \in \mathcal{G}} \hat{\mathcal{L}}_p(g)$. In a flexible or high-dimensional class, researchers may instead use cross-fitting, or leave-one-out prediction. For example, partition the observations into folds $I_1, \dots, I_V$, estimate $\hat{g}^{(-v)}$ without using outcomes in fold I_v , and set $\hat{g}_i = \hat{g}^{(-v)}(X_i)$, $i \in I_v$. The resulting direct score-adjustment estimator is

$$\begin{aligned} \hat{\tau}_{\text{dir}} &= \frac{1}{N_1} \sum_{i=1}^N Z_i \{Y_i - \hat{g}_i\} - \frac{1}{N_0} \sum_{i=1}^N (1 - Z_i) \{Y_i - \hat{g}_i\} \\ &= \hat{\tau}_{\text{DM}} - \Delta_Z(\hat{g}). \end{aligned} \tag{22}$$

Existing work on cross-estimation, leave-one-out adjustment, regularization, and direct machine-learning augmentation provides conditions under which estimated scores yield valid and efficient treatment-effect estimators (Bloniarz et al., 2016; Wager et al., 2016; Wu and Gagnon-Bartsch, 2018). We provide results on consistency and variance estimation in SI B. We note that Zhang and Ma (2019) develop a related approach that directly learns an augmentation function and derives the corresponding variance estimator from a semiparametric perspective. Their empirical loss uses an estimated, arm-centered influence function response, whereas our learner uses the raw-outcome complementary-weighted loss. These criteria have the same population target after translation and scale, but their finite-sample minimizers need not coincide.

4.2 Arm-Specific Outcome Regression

Instead of learning a single score directly, researchers may estimate the arm-specific outcome functions $m_z(x) \approx Y_i(z)$, and combine them as $g_{m_1, m_0}(x) = q m_1(x) + p m_0(x)$. If $m_1(X_i)$ predicts $Y_i(1)$ and $m_0(X_i)$ predicts $Y_i(0)$, then $g_{m_1, m_0}(X_i)$ predicts the ATE-relevant score

$$M_i = (1 - p)Y_i(1) + pY_i(0).$$

Given any pair of prediction rules m_1 and m_0 , define

$$\begin{aligned} \hat{\tau}(m_1, m_0) &= \frac{1}{N} \sum_{i=1}^N \left[m_1(X_i) - m_0(X_i) + \frac{Z_i}{p} \{Y_i - m_1(X_i)\} - \frac{1 - Z_i}{q} \{Y_i - m_0(X_i)\} \right] \\ &= \frac{1}{N} \sum_{i=1}^N \{m_1(X_i) - m_0(X_i)\} \\ &\quad + \frac{1}{N_1} \sum_{i=1}^N Z_i \{Y_i - m_1(X_i)\} \\ &\quad - \frac{1}{N_0} \sum_{i=1}^N (1 - Z_i) \{Y_i - m_0(X_i)\}. \end{aligned} \tag{23}$$

Because the treatment probability $p = N_1/N$ is known by design, Equation (23) is the familiar generalized-regression or AIPW estimator for a randomized experiment. The following algebraic identity connects this familiar estimator directly to the single-score framework.

Proposition 6 (AIPW is single-score adjustment). *For any numerical prediction vectors $\{m_1(X_i), m_0(X_i)\}_{i=1}^N$, including predictions estimated from the observed data,*

$$\hat{\tau}(m_1, m_0) = \hat{\tau}_{\text{DM}} - \Delta_Z \{(1 - p) m_1(X) + p m_0(X)\} = \hat{\tau}_{\text{DM}} - \Delta_Z \{g_{m_1, m_0}(X)\}. \tag{24}$$

Proposition 6 establishes that the proposed adjustment does not introduce a new treatment-effect estimator. Rather, it gives a new interpretation of familiar regression and augmentation estimators: they subtract the realized treatment–control imbalance in the estimated predictor of M .

In a low-dimensional setting, researchers may estimate m_1 and m_0 using separate regressions within the treatment and control groups. With linear arm-specific regressions on centered covariates, the resulting estimator is equivalent to fully interacted regression adjustment (Lin, 2013; Negi and Wooldridge, 2021). More generally, Equation (23) is the standard randomized-trial AIPW or generalized-regression estimator studied in the semiparametric literature (Robins et al., 1994, 1995; Tsiatis et al., 2008; Zhang et al., 2008). Flexible arm-specific regressions may be estimated using regularization or machine learning, with sample splitting, cross-fitting, or leave-one-out prediction used under the conditions developed in the corresponding literature (Bloniarz et al., 2016; Wager et al., 2016; Wu and Gagnon-Bartsch, 2018).

4.3 Discussion

The fixed-score theory clarifies the cost of learning the score. Conditional on $\widehat{g}$,

$$\begin{aligned} \text{MSE}_Z\{\widehat{\tau}(\widehat{g}) \mid \widehat{g}\} &= \frac{N}{N_1 N_0} S_{M-\widehat{g}}^2 \\ &= \underbrace{\frac{N}{N_1 N_0} S_{M-g^*}^2}_{\text{class-oracle risk}} + \underbrace{\frac{N}{N_1 N_0} \{S_{M-\widehat{g}}^2 - S_{M-g^*}^2\}}_{\text{excess score risk}}. \end{aligned} \quad (25)$$

The second term is nonnegative by the definition of g^* . Averaging it over repeated training samples gives the expected cost of learning the score rather than knowing the class oracle. The factor $\frac{N}{N_1 N_0} = \frac{1}{Npq}$ reflects the size and allocation of the target experiment, whereas the excess score risk depends on the training sample, the complexity of the candidate class, and the estimation procedure.

When $\widehat{g}$ is learned from the same experimental outcomes, it depends on the realized assignment, and the fixed-score decomposition in Equation (25) does not directly provide a finite-sample no-harm guarantee. Because our theory leads to the same covariate-adjustment estimators already established in the literature, we do not need to develop separate theory for each specific learner. The existing literature has already characterized the efficiency gains associated with these estimators (Lin, 2013; Negi and Wooldridge, 2021; Robins et al., 1994, 1995; Tsiatis et al., 2008; Zhang et al., 2008; Bloniarz et al., 2016; Wager et al., 2016; Wu and Gagnon-Bartsch, 2018; Zhang and Ma, 2019). Our contribution is to provide a new justification from a decision-theoretic perspective and to show that the resulting estimator is equivalent to direct adjustment for ex-post bias.

Accordingly, practical implementation does not require researchers to classify each pretreatment covariate separately as prognostic or nonprognostic. Researchers may prespecify a substantively reasonable set of baseline covariates and use them jointly to estimate the relevant score. In a modest low-dimensional setting, arm-specific linear regressions yield the usual fully interacted regression or AIPW estimator. When the candidate set is large or the prediction rule is flexible, regularization together with cross-fitting, or leave-one-out prediction should be used under the conditions established in the corresponding literature.

Which estimator should researchers use in practice? We compare their performance across a range of scenarios in the simulation section. One important lesson is that, because arm-specific estimators require estimating two separate nuisance functions, they can incur greater estimation costs and may therefore perform worse than our direct estimator in finite samples.

5 Simulation Design and Results

The simulations examine practical choice among adjustment strategies along two dimensions. The first is estimator performance across signal structures, allocation ratios, and sample sizes. With $N = 400$ and $K = 60$, the main study compares seven estimators across four scenarios. A focused extension holds $K = 60$ and balanced assignment fixed while varying $N \in \{160, 240, 400\}$ to compare the methods in smaller samples. The second dimension is covariate selection policy. We summarize this comparison at the end of the section, and Appendix D reports the full study.

5.1 Estimators

We begin with the difference-in-means estimator as the unadjusted benchmark, followed by three regression-adjustment estimators. All- X OLS represents the familiar additive regression adjustment using the full set of covariates, while All- X Lin allows the covariate slopes to differ across treatment arms. Because unpenalized adjustment may perform poorly when the number of covariates is large relative to the sample size ($K = 60$ and $N = 400$), One-step LASSO provides a regularized alternative to additive adjustment: it penalizes the covariate coefficients while leaving the treatment coefficient unpenalized.

The remaining estimators focus on the ATE-relevant prediction target. Following section 4, we consider direct and arm-specific estimators. Direct M -LASSO learns a single adjustment score under the complementary design-weighted loss. Arm-specific LASSO instead fits separate treatment- and control-arm outcome models and combines their predictions; the resulting estimator has the familiar cross-fitted AIPW form with the known assignment probability. Finally, the DGP oracle uses the known systematic score and therefore benchmarks performance in the absence of score-estimation error. Table 3 summarizes these seven estimators. Although we primarily use LASSO because of its computational convenience and familiarity to empirical researchers, in practice, researchers are free to use any machine-learning method. Implementation details are included in SI C.

Table 3: Estimators in the main simulation

Estimator	Construction	Reason for inclusion
Difference in means (DM)	No covariate adjustment.	No-adjustment benchmark.
All- X OLS	Additive OLS using all 60 prespecified covariates.	Familiar additive regression adjustment.
All- X Lin	Fully interacted Lin regression using all 60 covariates.	Familiar adjustment with arm-specific slopes.
One-step LASSO	One regression of Y on unpenalized Z and penalized X .	Regularized additive adjustment.
Direct M -LASSO	Cross-fitted learning under the complementary design-weighted loss.	Direct implementation of the paper’s score-learning target.
Arm-specific LASSO (AIPW)	Cross-fitted arm regressions combined using the design proportions.	Familiar cross-fitted AIPW implementation.
DGP oracle	Adjustment using the known systematic score.	Benchmark without score-estimation error.

5.2 Data-generating framework

To evaluate these estimators in a common setting, each baseline replication generates a finite-population science table for $N = 400$ units and $K = 60$ pretreatment covariates. Covariates are

independent across units and follow

$$X_i \sim \mathcal{N}(0, \Sigma), \text{ where } \Sigma_{jk} = 0.3^{|j-k|}, i = 1, \dots, N. \quad (26)$$

Thus, every covariate has superpopulation variance one, and correlations decline with the distance between their indices. We use the prespecified covariance matrix Σ for all normalizations; the realized finite population is not recentered or rescaled.

We generate the potential outcomes through the unit-level treatment effect τ_i and the score M_i from Proposition 3. For scenario s , let $g_s^\circ(X)$ be a normalized covariate index that carries useful signal for M , and let $h_s^\circ(X)$ be a normalized index that generates treatment-effect heterogeneity. With $p = N_1/N$, $q = 1 - p$, and $\varepsilon_i \sim \mathcal{N}(0, 1)$ independently of X_i , define

$$\begin{aligned} M_i &= \sqrt{R_M^2} g_s^\circ(X_i) + \sqrt{1 - R_M^2} \varepsilon_i, \\ \tau_i &= 0.20 + \sigma_{\tau,s} h_s^\circ(X_i), \\ Y_i(1) &= M_i + p\tau_i, \text{ and } Y_i(0) = M_i - q\tau_i. \end{aligned} \quad (27)$$

This construction guarantees both $Y_i(1) - Y_i(0) = \tau_i$ and $qY_i(1) + pY_i(0) = M_i$.

The parameter R_M^2 controls how much of the relevant score can be predicted from baseline covariates. Because every nonzero $g_s^\circ(X)$ is normalized to have superpopulation variance one and is independent of ε_i ,

$$\mathbb{E}(M_i | X_i) = \sqrt{R_M^2} g_s^\circ(X_i), \text{ and } R_M^2 = \frac{\text{Var}\{\mathbb{E}(M_i | X_i)\}}{\text{Var}(M_i)}. \quad (28)$$

Thus, $R_M^2 = 0$ means that the covariates contain no systematic information about ATE estimation error, whereas $R_M^2 = 0.50$ means that they explain one half of the superpopulation variation in M . It is neither a measure of realized covariate balance nor an R^2 for one treatment arm. Because the generated finite population is not restandardized, its empirical predictive fit fluctuates around the nominal value.

Table 4 summarizes how the four scenarios vary the predictive signal, treatment-effect heterogeneity, and allocation ratio. Its final column explains why each scenario is included.

Table 4: Structural scenarios in the main simulation

Scenario	p	R_M^2	$\sigma_{\tau,s}$	Purpose
Null	1/2	0	0	Compares the methods when no baseline covariate can improve ATE precision.
Sparse M signal	1/2	0.50	0	Tests whether the methods recover a gain carried by four of 60 covariates.
Arm-specific cancellation	1/2	0	2	Makes outcomes predictable within arms even though the relevant score has no covariate signal.
Unequal allocation with heterogeneous effects	1/4	0.50	1	Makes the arm weights consequential while different covariates predict the score and treatment effects.

To implement the signal and heterogeneity components, we use the following sparse normalized indices:

$$\begin{aligned}
g_s^\circ(X) &= \frac{X_1 + X_2 + X_3 + X_4}{\sqrt{6.214}} && \text{in the sparse-signal scenario,} \\
h_s^\circ(X) &= \frac{X_1 + X_2 - X_3}{\sqrt{2.82}} && \text{in the cancellation scenario,} \\
g_s^\circ(X) &= \frac{X_1 + X_2 + X_3}{\sqrt{4.38}}, \quad h_s^\circ(X) = \frac{X_4 - X_5 + X_6}{\sqrt{1.98}} && \text{under unequal allocation.} \quad (29)
\end{aligned}$$

The denominators are the theoretical standard deviations of the displayed indices under Σ . An index not specified for a scenario is zero.

After generating the science table, we assign exactly $N_1 = pN$ units to treatment by complete randomization. In replication r , all errors are evaluated against the realized finite-population ATE. When effects are heterogeneous, τ_r need not equal 0.20, although its expectation across generated populations is 0.20.

5.3 Performance measures

We evaluate the estimators at two levels: ATE performance for all methods and held-out score-prediction performance for the cross-fitted learners. For each estimator ℓ , we report bias, root MSE, and MSE relative to the difference in means:

$$\text{RelativeMSE}_\ell = \frac{\sum_{r=1}^R (\hat{\tau}_{\ell r} - \tau_r)^2}{\sum_{r=1}^R (\hat{\tau}_{\text{DM}, r} - \tau_r)^2}. \quad (30)$$

A value below one indicates an MSE improvement over no adjustment. All methods within a replication use the same science table and assignment. We report paired Monte Carlo standard errors (MCSE) and 95% Monte Carlo intervals for relative MSE.

For Direct and Arm-specific LASSO, we additionally evaluate the held-out score predictions. The reported “ M explained” measure is one minus the fold-centered prediction risk of $\hat{g}$ relative to a constant score; we also report the fold-centered correlation with M and the improvement in the observable held-out complementary weighted loss. These are honest out-of-fold diagnostics. We do not compare them with in-sample losses for All- X OLS, All- X Lin, or One-step LASSO.

To compare performance across sample sizes, we summarize each method’s distance from the known-score benchmark using the normalized oracle gap

$$G_{\ell, N} = \frac{\text{MSE}(\hat{\tau}_{\ell, N}) - \text{MSE}(\hat{\tau}_{\text{or}, N})}{\text{MSE}(\hat{\tau}_{\text{DM}, N})} = \text{RelativeMSE}_{\ell, N} - \text{RelativeMSE}_{\text{or}, N}. \quad (31)$$

Thus, $G_{\ell, N}$ is the MSE difference from the DGP oracle in units of difference-in-means MSE. It is zero for the oracle, and smaller values indicate performance closer to that benchmark. In the cancellation scenario, the oracle equals the difference in means, so $G_{\ell, N}$ is the proportional MSE difference between feasible adjustment and no adjustment. We report paired Monte Carlo standard errors and untruncated 95% Monte Carlo intervals for this measure.

5.4 Results

5.4.1 Baseline performance

Table 5 and Figure 1 first report baseline ATE performance. After interpreting those results, we use Table 6 to link the performance of the cross-fitted procedures to their held-out score predictions. In this baseline study, bias is not distinguishable from Monte Carlo noise: the largest absolute bias is 0.0056 with an MCSE of 0.0035, and no bias estimate exceeds 1.61 MCSEs in absolute value.

Table 5: ATE performance in the baseline simulation

Scenario	Estimator	Bias (MCSE)	RMSE	Rel. MSE (MCSE) [95% MCI]
Null	DM	-0.0007 (0.0033)	0.1033	1.000 (0.000) [1.000, 1.000]
Null	All-X OLS	-0.0006 (0.0035)	0.1118	1.173 (0.029) [1.116, 1.229]
Null	All-X Lin	-0.0016 (0.0036)	0.1136	1.210 (0.034) [1.143, 1.277]
Null	One-step LASSO	-0.0009 (0.0033)	0.1034	1.002 (0.003) [0.996, 1.008]
Null	Direct M-LASSO	-0.0008 (0.0033)	0.1033	1.001 (0.003) [0.996, 1.006]
Null	Arm-specific LASSO (AIPW)	-0.0006 (0.0033)	0.1033	1.001 (0.003) [0.995, 1.007]
Null	DGP oracle	-0.0007 (0.0033)	0.1033	1.000 (0.000) [1.000, 1.000]
Sparse M signal	DM	-0.0009 (0.0033)	0.1038	1.000 (0.000) [1.000, 1.000]
Sparse M signal	All-X OLS	-0.0010 (0.0024)	0.0767	0.546 (0.026) [0.495, 0.596]
Sparse M signal	All-X Lin	-0.0008 (0.0025)	0.0779	0.563 (0.027) [0.511, 0.616]
Sparse M signal	One-step LASSO	-0.0010 (0.0023)	0.0736	0.502 (0.020) [0.463, 0.540]
Sparse M signal	Direct M-LASSO	-0.0008 (0.0024)	0.0743	0.512 (0.019) [0.474, 0.550]
Sparse M signal	Arm-specific LASSO (AIPW)	-0.0007 (0.0024)	0.0756	0.531 (0.019) [0.494, 0.567]
Sparse M signal	DGP oracle	-0.0006 (0.0023)	0.0723	0.485 (0.022) [0.442, 0.527]
Arm-specific cancellation	DM	-0.0033 (0.0031)	0.0991	1.000 (0.000) [1.000, 1.000]
Arm-specific cancellation	All-X OLS	-0.0044 (0.0037)	0.1165	1.381 (0.044) [1.296, 1.467]
Arm-specific cancellation	All-X Lin	-0.0056 (0.0035)	0.1111	1.256 (0.033) [1.191, 1.321]
Arm-specific cancellation	One-step LASSO	-0.0035 (0.0031)	0.0996	1.009 (0.004) [1.001, 1.018]
Arm-specific cancellation	Direct M-LASSO	-0.0032 (0.0031)	0.0993	1.004 (0.004) [0.997, 1.012]
Arm-specific cancellation	Arm-specific LASSO (AIPW)	-0.0039 (0.0032)	0.1011	1.041 (0.010) [1.020, 1.061]
Arm-specific cancellation	DGP oracle	-0.0033 (0.0031)	0.0991	1.000 (0.000) [1.000, 1.000]
Unequal allocation with HTE	DM	-0.0012 (0.0037)	0.1185	1.000 (0.000) [1.000, 1.000]
Unequal allocation with HTE	All-X OLS	0.0017 (0.0034)	0.1064	0.806 (0.042) [0.724, 0.889]
Unequal allocation with HTE	All-X Lin	-0.0021 (0.0035)	0.1114	0.884 (0.050) [0.786, 0.983]
Unequal allocation with HTE	One-step LASSO	-0.0012 (0.0030)	0.0940	0.629 (0.029) [0.573, 0.685]
Unequal allocation with HTE	Direct M-LASSO	-0.0028 (0.0026)	0.0834	0.495 (0.018) [0.459, 0.532]
Unequal allocation with HTE	Arm-specific LASSO (AIPW)	-0.0018 (0.0027)	0.0865	0.533 (0.020) [0.494, 0.572]
Unequal allocation with HTE	DGP oracle	-0.0025 (0.0025)	0.0789	0.443 (0.022) [0.401, 0.486]

Notes: Bias is calculated relative to the realized finite-population ATE τ_r . RMSE is on the scale of the outcome. Each relative-MSE entry uses the difference in means as its within-scenario benchmark; parentheses contain paired Monte Carlo standard errors (MCSE), and brackets contain 95% Monte Carlo intervals.

The null and sparse-signal scenarios first show how regularization performs under balanced allocation. In the null scenario, One-step LASSO has relative MSE 1.002, and Direct and Arm-specific LASSO both round to 1.001. By contrast, All-X OLS and Lin increase MSE by 17.3% and 21.0%. When the useful signal is sparse, One-step LASSO attains relative MSE 0.502, compared with 0.512 for Direct LASSO, 0.531 for Arm-specific LASSO/AIPW, and 0.485 for the oracle.

Arm-specific cancellation gives the regularized methods a harder no-signal case. All-X OLS and Lin raise relative MSE to 1.381 and 1.256, respectively. One-step and Direct LASSO remain

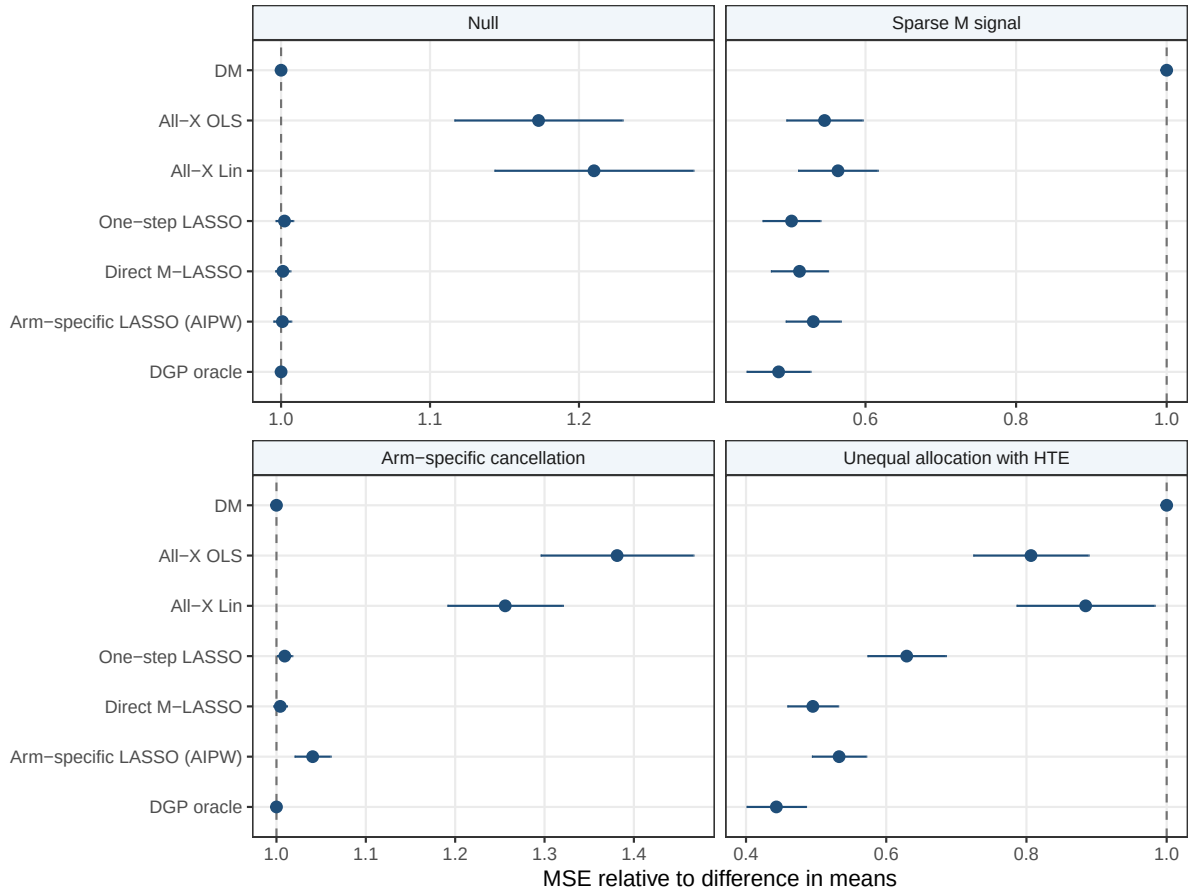


Figure 1: MSE relative to the difference in means

Notes: Points show relative MSE, and horizontal intervals equal the estimate plus or minus 1.96 paired Monte Carlo standard errors. The dashed vertical line marks the MSE of the unadjusted difference in means.

close to no adjustment, at 1.009 and 1.004. Arm-specific LASSO/AIPW has relative MSE 1.041 because two strong arm-specific relationships must cancel after being estimated separately.

Unequal allocation makes the choice of prediction target consequential. Direct LASSO attains relative MSE 0.495, and Arm-specific LASSO/AIPW attains 0.533, compared with 0.629 for conventional One-step LASSO. The oracle value is 0.443, while All- X OLS and Lin are farther away at 0.806 and 0.884. This pattern is consistent with the value of complementary design weighting when allocation is unequal.

Table 6: Held-out score-prediction performance

Scenario	Learner	M explained	Correlation	Loss improvement
Null	Direct M-LASSO	-0.002	-0.010	-0.002
Null	Arm-specific LASSO (AIPW)	-0.002	-0.008	-0.002
Null	DGP oracle	0.000	–	0.003
Sparse M signal	Direct M-LASSO	0.475	0.695	0.471
Sparse M signal	Arm-specific LASSO (AIPW)	0.461	0.692	0.457
Sparse M signal	DGP oracle	0.500	0.708	0.497
Arm-specific cancellation	Direct M-LASSO	-0.004	-0.003	-0.002
Arm-specific cancellation	Arm-specific LASSO (AIPW)	-0.024	-0.006	-0.010
Arm-specific cancellation	DGP oracle	0.000	–	0.003
Unequal allocation with HTE	Direct M-LASSO	0.443	0.677	0.370
Unequal allocation with HTE	Arm-specific LASSO (AIPW)	0.415	0.651	0.348
Unequal allocation with HTE	DGP oracle	0.500	0.708	0.423

Notes: Entries average replication-level diagnostics. M explained is one minus fold-centered prediction risk. Correlation uses fold-centered held-out predictions. Loss improvement is measured against a held-out intercept-only learner under the complementary design-weighted loss. Correlation is undefined when the oracle score is constant.

The prediction diagnostics explain these MSE patterns. In the sparse-signal scenario, Direct and Arm-specific LASSO explain 0.475 and 0.461 of the held-out variation in M , compared with 0.500 for the oracle. Under unequal allocation, the corresponding values are 0.443, 0.415, and 0.500. In the null and cancellation scenarios, the feasible learners’ explained fractions are zero or slightly negative, correctly indicating that fitting a score offers no systematic gain. Direct LASSO is more accurate than the arm-specific route in the two signal settings considered here, plausibly because each direct fit uses the entire outer training sample, whereas the arm-specific fits divide those observations between treatment arms.

5.4.2 Performance across sample sizes

The baseline results at $N = 400$ do not show whether the relative performance of the estimators changes in smaller experiments. We therefore hold $K = 60$ and $p = 1/2$ fixed and repeat the sparse-signal and arm-specific-cancellation scenarios at $N \in \{160, 240, 400\}$. All seven estimators retain the fitting rules described above, and the $N = 400$ results are reused from the baseline study. The smallest sample is deliberately demanding: each arm contains 80 observations, so All- X Lin estimates 60 slopes and an intercept with only 19 residual degrees of freedom per arm. The sparse-signal scenario compares how well the methods convert useful covariate information into precision, whereas cancellation compares their behavior when covariates predict outcomes within arms but

the oracle coincides with no adjustment.

Figure 2 displays the normalized oracle gap defined in Equation (31), and Table A.2 reports exact results for all seven estimators.

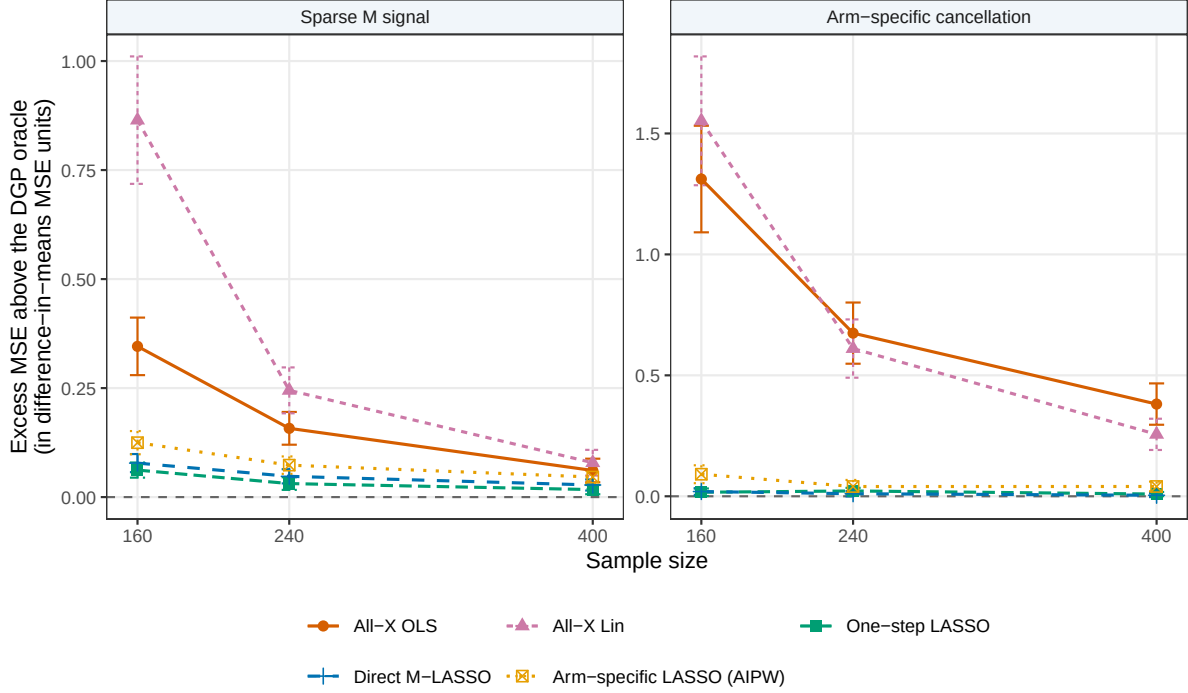


Figure 2: Estimator performance relative to the DGP oracle across sample sizes

Notes: Points report the normalized oracle gap $G_{\ell,N}$ from Equation (31); vertical intervals equal the estimate plus or minus 1.96 paired Monte Carlo standard errors. Lower values indicate performance closer to the DGP oracle. Both panels hold $K = 60$ and $p = 1/2$ fixed. The DGP oracle, whose gap is zero, and the difference in means are omitted for legibility. The $N = 400$ estimates reuse the baseline simulation.

In the sparse-signal setting, the methods separate more sharply as the sample becomes smaller. At $N = 160$, the oracle gap is 0.346 for All- X OLS and 0.865 for All- X Lin. The corresponding gaps are 0.062 for One-step LASSO, 0.078 for Direct M -LASSO, and 0.125 for Arm-specific LASSO/AIPW. Thus, the regularized methods remain much closer to the oracle in the smallest sample, whereas fully interacted regression loses the available precision gain and performs worse than no adjustment.

Cancellation sharpens the practical comparison. At $N = 160$, All- X OLS and Lin lie 131.2% and 155.2% above the oracle, respectively. One-step and Direct LASSO lie only 1.7% and 2.0% above it, whereas Arm-specific LASSO/AIPW lies 9.1% above it. The arm-specific gap is approximately 4% at both $N = 240$ and $N = 400$; at those sample sizes, Direct LASSO remains within about 1% of the oracle. A plausible finite-sample explanation is that each arm-specific nuisance regression is learned from only one treatment arm, so estimation errors need not cancel even when the underlying relationships do. This pattern does not conflict with semiparametric efficiency results for AIPW under their regularity and nuisance-estimation conditions, and it does not establish uniform dominance by the direct learner.

5.4.3 Covariate selection policies

To avoid additional inferential complications from reusing target outcomes, Appendix D uses an independent selection sample for outcome-dependent covariate selection and for the common external-score refit. It compares five covariate-set rules: control-outcome prediction, covariate imbalance, their intersection, their union, and selection under the design-relevant weighted loss. The covariate imbalance rule uses assignments and covariates but never target outcomes.

The sparse-signal scenario provides a benchmark: the control-outcome and design-relevant rules both attain external-score relative MSE 0.506. Under arm-specific cancellation, the control-outcome rule retains variables whose arm-specific contributions cancel in M and raises relative MSE to 1.022, whereas the design-relevant rule selects essentially no covariates and reproduces no adjustment. Under unequal allocation, the corresponding relative MSEs are 0.545 and 0.535. In these scenarios, the balance-based rules are less reliable, and Lin regression preserves the same qualitative ranking. Table A.1 and Figure A.1 report the complete results. The study therefore establishes covariate imbalance should not be used in practice. Although control-outcome prognosticity is better than covariate imbalance policy, it is not sufficient for prognosticity with respect to ATE estimation error.

5.5 Practical guidance for estimator choice

If a single procedure must be prespecified for use across different treatment allocation ratios, Direct M -LASSO exhibits the strongest overall performance in our simulations. It performs best among the feasible estimators under unequal allocation and performs nearly as well as One-step LASSO under equal allocation.

The arm-specific AIPW estimator remains a credible alternative when separate outcome models are substantively desirable, but it can trail the direct learner when each arm provides limited training data. With $K = 60$, unpenalized All- X OLS and Lin are not attractive default choices. More generally, these results are scenario-specific performance comparisons, not a proposition of uniform dominance.

6 Discussion

Discussions of covariate adjustment in randomized experiments routinely combine two questions. First, which pretreatment variables are *prognostic*? Second, which pretreatment variables are *imbalanced* in the realized assignment? Conventional approaches call covariates prognostic when they predict outcomes and imbalanced when their distributions differ by treatment assignment. However, our framework shows that both questions require an ATE-specific answer. The relevant pair of oracle objects is

$$\underbrace{M_i = (1 - p)Y_i(1) + pY_i(0)}_{\text{the object to predict}} \quad \text{and} \quad \underbrace{\Delta_z(M)}_{\text{the imbalance that equals realized ATE error}}, \quad (32)$$

where lowercase z denotes the assignment that was actually realized.

The theory above shows that reducing randomization variance requires predicting and remov-

ing $\Delta_z(M)$. The relevant prognostic target is therefore M , rather than Y in a generic sense. Example 1 demonstrates that even perfect arm-specific outcome prediction by X need not reduce the MSE of the ATE when the two associations cancel in M . Likewise, raw covariate imbalance $\Delta_z(X)$ does not by itself determine the bias; the exact realized error is $\Delta_z(M)$.

The oracle pair in (32) is not observed because the two potential outcomes are never jointly observed. The feasible counterparts are therefore

$$\underbrace{g(X_i)}_{\text{a pretreatment predictor of } M_i} \quad \text{and} \quad \underbrace{\Delta_z\{g(X)\}}_{\text{the predicted component of realized ATE error}}. \quad (33)$$

The score g should be chosen according to how well it predicts M , up to an additive constant. Once the score has been chosen, its signed realized imbalance $\Delta_z\{g(X)\}$ is the point correction. This distinction is the central message of this section: *prognosticity selects the score, while score imbalance determines the realized correction.*

6.1 Balance Tables

Conventional balance tables report treated and control means, standardized mean differences, and sometimes balance-test p -values for each baseline covariate. These quantities are useful for describing the realized assignment, checking implementation, and documenting the design. Our theory implies, however, that they do not provide a coherent rule for selecting an ATE adjustment model. A balance statistic measures how differently a variable was distributed across treatment arms; it does not measure whether the variable predicts the ex-post bias. Conversely, a measure of predictive performance is an ex-ante property of a score and does not describe how much that score will change the estimate in the particular assignment observed.

A preliminary balance test classifies assignments according to whether a raw covariate difference is statistically unusual under randomization. Table 7 shows the resulting four cases. When a useful prognostic variable happens not to cross the balance-test threshold, the rule withholds an adjustment that has ex-ante value. When an irrelevant variable happens to cross the threshold, the rule forces an unnecessary coefficient into the analysis. The two remaining cells produce reasonable decisions only by coincidence. The significance threshold is therefore neither necessary nor sufficient for choosing an adjustment score.

The problem is sharpened when balance is judged by a p -value. A p -value combines the realized mean difference with the randomization variance of the covariate. It indicates how surprising the raw imbalance is under the design, not how large the implied ATE correction is in outcome units. A high-variance, weakly prognostic covariate can have a small p -value, while a highly prognostic score can have a large p -value because it happened to be close to balanced. Neither event changes the score's out-of-sample ability to predict M .

6.2 A Recommended Prognostic Balance Report

We recommend retaining the conventional raw-covariate balance table and adding an outcome- and estimand-specific prognostic-score panel. The conventional panel documents the assignment and can reveal coding mistakes, attrition, deviation from the randomization protocol, or differences

Table 7: The four cases generated by a balance-test-and-adjust rule

	Balance test does not reject	Balance test rejects
The covariate contributes to an ATE-relevant prognostic score	<i>False negative for the adjustment decision.</i> The rule omits useful information because the covariate happened to be relatively balanced in this assignment. The realized point correction may be small, but the score can still improve ex-ante precision.	<i>Adjustment for the right variable, but for the wrong reason.</i> The covariate is useful because it predicts M , not because a preliminary test crossed a threshold. The same score should be available in assignments in which the test does not reject.
The covariate has no incremental predictive value for M	<i>No adjustment, but only by coincidence.</i> The correct oracle coefficient is zero; the balance-test result adds no information about ATE error.	<i>False positive for the adjustment decision.</i> The imbalance may be unusual, but it does not predict ATE error. Estimating and forcing an unnecessary coefficient can increase finite-sample MSE.

that readers regard as substantively important. It should not be used as a preliminary-test model-selection device. The prognostic panel connects baseline information to the estimation target.

Let $\hat{g}_i$ denote a prespecified, externally trained, or cross-fitted prediction of M_i . For each outcome and estimand, report

$$\bar{\hat{g}}_1 = \frac{1}{N_1} \sum_{i=1}^N Z_i \hat{g}_i, \quad \bar{\hat{g}}_0 = \frac{1}{N_0} \sum_{i=1}^N (1 - Z_i) \hat{g}_i, \quad (34)$$

$$\Delta_Z(\hat{g}) = \bar{\hat{g}}_1 - \bar{\hat{g}}_0, \quad D_{\hat{g}} = \frac{\Delta_Z(\hat{g})}{\sqrt{N/(N_1 N_0)} S_{\hat{g}}}. \quad (35)$$

The unstandardized difference $\Delta_Z(\hat{g})$ is of primary interest because it indicates the point correction and is measured in outcome units. The standardized difference $D_{\hat{g}}$, together with an optional randomization p -value, answers the separate descriptive question of how unusual the score imbalance is under the design.

The resulting division of labor is as follows. The prognostic balance panel reports how that useful score was distributed in the realized assignment. A small score imbalance means that the adjustment changes the point estimate little in this experiment. It does not imply that the score should have been excluded. Similarly, a large raw-covariate imbalance may be noteworthy (and worth disclosing for the sake of transparency) without necessitating a change in the set of covariates used for adjustment.

6.3 Relationship to the Literature

The framework connects four related strands of work: covariate adjustment, prognostic scores, conditional inference after imbalance, and design-stage balance. This subsection clarifies how the paper draws on each strand and where its decision-theoretic interpretation differs.

Covariate Adjustment. Covariate adjustment can be implemented in many ways. Except those mentioned in the section 4, a growing literature uses regularization and machine learning to estimate outcome regressions or augmentation functions when the covariate vector is large. [Bloniarz et al. \(2016\)](#) develop LASSO adjustment under the finite-population potential-outcomes framework. [Wager et al. \(2016\)](#) show that a broad class of risk-consistent prediction methods can be combined with cross-estimation to obtain valid and efficient treatment-effect estimators. LOOP uses leave-one-out prediction for a similar purpose ([Wu and Gagnon-Bartsch, 2018](#)), while [Zhang and Ma \(2019\)](#) construct a direct super learner for the optimal augmentation.

These papers primarily address feasible estimation and inference when the prediction rule is learned from observed outcomes. Our oracle results address a logically prior question: what should the learner predict in the first place? Guided by this framework, we derive both direct and arm-specific estimators and show that the latter is equivalent to a broad class of existing covariate-adjustment estimators in the literature. This provides a new theoretical foundation for existing covariate-adjustment estimators.

Prognostic Scores. Our notion of a design-relevant prognostic score is related to, but distinct from, the prognostic score introduced by [Hansen \(2008\)](#) and the prognosis-weighted balance tests developed by [Bicalho et al. \(2026\)](#). Hansen defines a prognostic score ($\Psi_0(X)$) as a dimension-reducing balancing score for the control potential outcome, typically satisfying $Y(0) \perp X \mid \Psi_0(X)$. This is a conditional-independence, or sufficiency, property intended primarily to facilitate matching, stratification, and confounding adjustment in observational studies.

By contrast, [Bicalho et al. \(2026\)](#) build on this idea for a different inferential objective. They estimate the control potential outcome from control-group observations and use the treatment–control difference in fitted control outcomes—equivalently, a prognosis-weighted combination of covariate imbalances—as an omnibus test of as-if random assignment or continuity of potential outcomes.

Conditional Imbalance and Post-Assignment Decisions. Our decision proposition is most closely related to the conditional-inference work of [Zhang et al. \(2019\)](#) and [Johansson and Nordin \(2022\)](#). Those papers establish that the difference-in-means estimator can be conditionally biased given observed covariate imbalance and develop adjusted estimation or inference procedures to address it. Their results are important precedents for our concept of ex-post bias. We do not aim to re-derive the conditional-inference results; rather, we develop a framework for covariate-adjustment decisions that recovers traditional estimators.

Design-Stage Balance. Covariates can also be used before treatment assignment. Blocking, matching, stratification, and rerandomization restrict the assignment mechanism so that large imbalances in important baseline variables are less likely or impossible. [Morgan and Rubin \(2012\)](#) formalize rerandomization based on a prespecified balance criterion. [Bai \(2022\)](#) shows that, within a broad class of stratified designs with equal treatment probabilities, an appropriately constructed matched-pair design can maximize precision; in an important special case, the optimal match is based on a baseline outcome. [Li and Ding \(2020\)](#) further show how rerandomization and regression adjustment can be combined, since design-stage and analysis-stage uses of covariates provide complementary efficiency gains.

Our framework is complementary to design-stage balancing. Design methods reduce the ex-ante variation in prognostic imbalance by restricting the assignments that can occur. Our problem begins after an assignment has been realized and asks what the resulting pretreatment information reveals about the error of the unadjusted estimator.

6.4 Practical Guidance

Researchers should collect a rich set of pretreatment covariates whenever feasible, prioritizing information that can help predict the design-relevant score M . The decomposition $M_i = Y_i(0) + (1 - p)\tau_i$ provides a useful guide: relevant covariates may predict baseline potential outcomes, treatment-effect heterogeneity, or both. The objective, however, is to predict their weighted combination, rather than either component in isolation, because their associations with a covariate may reinforce or offset one another. Collecting a broad set of candidate predictors therefore provides useful flexibility, without requiring that every collected variable enter an unregularized adjustment model.

Because M is unobserved, researchers estimate its predictor $g(X)$ using either approach developed above: direct or arm-specific estimation. With many covariates or flexible learners, regularization and cross-fitting can help control overfitting and the cost of estimating the score. Under either approach, adjustment subtracts the treatment–control difference in the fitted score from the unadjusted estimate: $\hat{\tau}_{\text{adj}} = \hat{\tau}_{\text{DM}} - \Delta_Z(\hat{g})$.

No preliminary balance test is needed to decide whether to adjust or which covariates to include. These decisions should be guided by prediction of M , rather than the statistical significance of realized covariate differences. The adjustment already incorporates realized balance through $\Delta_Z(\hat{g})$: when the fitted score is nearly balanced, the point correction is correspondingly small, without requiring a separate testing step. Conventional balance tables may still be reported for transparency and to assess experimental implementation, but they should not serve as a prerequisite for adjustment or as a covariate-selection rule.

7 Conclusion

This paper develops a finite-population framework for deciding when and how to adjust for pretreatment covariates after a randomized experiment has been realized. The framework reconciles the three debates in the literature and practice. Ex-post error correction and ex-ante precision improvement are therefore not competing rationales for adjustment; they are two interpretations of the same projection. From this framework, we derive a direct estimator as well as an arm-specific estimator, with the latter recovering the familiar generalized-regression and AIPW estimators. The central argument is that constructing or selecting an adjustment score, prognosticity for M is the sole criterion. Raw covariate imbalance, balance-test p -values, generic prediction of the observed outcome, and prediction of one potential outcome are not independent adjustment criteria; they are informative only when they help predict M .

The present analysis focuses on the finite-population ATE under complete randomization. Extending the decision framework to stratified, clustered, rerandomized, multi-arm, or repeated-experiment settings, and to estimands other than the ATE, requires deriving the corresponding

design- and estimand-specific error score. Further work can also sharpen finite-sample guarantees for data-adaptive score learning. The broader lesson, however, is already clear: covariate adjustment should be guided not by which baseline variables happen to be imbalanced, but by which pretreatment information predicts the randomization error relevant to the estimand. Randomization validates the unadjusted estimator across assignments; prognostic adjustment uses baseline information to improve the estimate from the assignment that was actually observed.

Statement of AI Usage

In preparing this paper, the authors used ChatGPT Sol models for the following tasks: constructing illustrative examples; assisting with code for generating figures and tables from simulation results; drafting and refining descriptions of the simulation design and results; and checking mathematical proofs. All AI-assisted outputs were carefully reviewed, and where necessary, revised by the authors. The authors take full responsibility for the content of the paper.

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Supplementary Information

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A Proofs

We define $q = 1 - p$ throughout the proof.

A.1 Proof of Proposition 1

Proof. Because a is $\mathcal{I}$ -measurable, it is constant after conditioning on $\mathcal{I}$. Hence

$$\mathbb{E}_Z(e_0 - a \mid \mathcal{I}) = \mathbb{E}_Z(e_0 \mid \mathcal{I}) - a = B_{\mathcal{I}} - a.$$

and

$$\text{Var}_Z(e_0 - a \mid \mathcal{I}) = \text{Var}_Z(e_0 \mid \mathcal{I}).$$

Define conditional squared-error risk $\mathcal{R}_{\mathcal{I}}(a) = \mathbb{E}_Z[\{\hat{\tau}_a - \tau\}^2 \mid \mathcal{I}]$. Applying the conditional bias–variance decomposition gives

$$\mathcal{R}_{\mathcal{I}}(a) = \text{Var}_Z(e_0 \mid \mathcal{I}) + \{B_{\mathcal{I}} - a\}^2,$$

and

$$\mathcal{R}_{\mathcal{I}}(0) = \text{Var}_Z(e_0 \mid \mathcal{I}) + \{B_{\mathcal{I}}\}^2,$$

Thus, $\mathcal{R}_{\mathcal{I}}(0) - \mathcal{R}_{\mathcal{I}}(a) = B_{\mathcal{I}}^2 - [B_{\mathcal{I}} - a(\mathcal{I})]^2$. Averaging the conditional result gives the unconditional results. Because the variance term does not depend on a , the squared-bias term is uniquely minimized by $a^* = B_{\mathcal{I}}$. $\square$

A.2 Proof of Proposition 2

Proof. The tower property gives

$$\mathbb{E}_Z(B_2 \mid \mathcal{I}_1) = \mathbb{E}_Z\{\mathbb{E}_Z(e_0 \mid \mathcal{I}_2) \mid \mathcal{I}_1\} = \mathbb{E}_Z(e_0 \mid \mathcal{I}_1) = B_1.$$

The decomposition $e_0 - B_1 = (e_0 - B_2) + (B_2 - B_1)$ is orthogonal in L^2 , because $e_0 - B_2$ is orthogonal to every $\mathcal{I}_2$ -measurable variable. Squaring and taking expectations yields the risk identity. The variance identity follows from the law of total variance applied to B_2 conditional on $\mathcal{I}_1$. $\square$

A.3 Proof of Proposition 3

Proof. Because $\bar{A} = p\bar{A}_1 + q\bar{A}_0$, we have $\bar{A}_1 - \bar{A} = q(\bar{A}_1 - \bar{A}_0) = q\Delta_Z(A)$,

Starting from the difference-in-means estimator,

$$\hat{\tau}_{\text{DM}} - \tau = \{\bar{Y}_1(1) - \bar{Y}(1)\} - \{\bar{Y}_0(0) - \bar{Y}(0)\}.$$

The identities

$$\bar{A}_1 - \bar{A} = q\Delta_Z(A), \quad \bar{A}_0 - \bar{A} = -p\Delta_Z(A)$$

give

$$\begin{aligned} \hat{\tau}_{\text{DM}} - \tau &= q\Delta_Z\{Y(1)\} + p\Delta_Z\{Y(0)\} \\ &= \Delta_Z\{qY(1) + pY(0)\} \\ &= \Delta_Z(M). \end{aligned}$$

□

A.4 Proof of Proposition 4

Proof. Throughout the proof, each candidate score $g \in \mathcal{G}$ is held fixed with respect to the treatment assignment. Write $g_i = g(X_i)$, $R_i(g) = M_i - g_i$, and $\bar{g} = \frac{1}{N} \sum_{i=1}^N g_i$.

For any fixed scalar finite-population variable $A = (A_1, \dots, A_N)$, the treated units form a simple random sample without replacement of size N_1 . Therefore, $\mathbb{E}_Z(\bar{A}_1) = \bar{A}$ and $\text{Var}_Z(\bar{A}_1) = \left(\frac{1}{N_1} - \frac{1}{N}\right) S_A^2 = \frac{N_0}{NN_1} S_A^2$. Moreover, $\bar{A}_0 = \frac{N\bar{A} - N_1\bar{A}_1}{N_0}$, so $\Delta_Z(A) = \bar{A}_1 - \bar{A}_0 = \frac{N}{N_0} (\bar{A}_1 - \bar{A})$. It follows that

$$\mathbb{E}_Z\{\Delta_Z(A)\} = 0 \tag{36}$$

and

$$\text{Var}_Z\{\Delta_Z(A)\} = \frac{N}{N_1 N_0} S_A^2. \tag{37}$$

By Proposition 3 and the linearity of Δ_Z ,

$$\begin{aligned} \hat{\tau}(g) - \tau &= \hat{\tau}_{\text{DM}} - \tau - \Delta_Z\{g(X)\} \\ &= \Delta_Z(M) - \Delta_Z\{g(X)\} \\ &= \Delta_Z\{M - g(X)\} \\ &= \Delta_Z\{R(g)\}. \end{aligned}$$

Applying (36) and (37) to the fixed residual $R(g)$ gives

$$\begin{aligned}\text{MSE}_Z\{\hat{\tau}(g)\} &= \mathbb{E}_Z [\{\Delta_Z(M) - \Delta_Z\{g(X)\}\}^2] \\ &= \text{Var}_Z [\Delta_Z\{M - g(X)\}] \\ &= \frac{N}{N_1 N_0} S_{M-g}^2.\end{aligned}\tag{38}$$

Because $N/(N_1 N_0) > 0$ does not depend on g ,

$$\arg \min_{g \in \mathcal{G}} \text{MSE}_Z\{\hat{\tau}(g)\} = \arg \min_{g \in \mathcal{G}} \mathbb{E}_Z [\{\Delta_Z(M) - \Delta_Z\{g(X)\}\}^2] = \arg \min_{g \in \mathcal{G}} S_{M-g}^2.\tag{39}$$

Because $g(X)$ is constant within each observed X -profile, $\Delta_Z\{g(X)\}$ is measurable with respect to $\mathcal{F}_X$. More explicitly, if $x^{(1)}, \dots, x^{(J)}$ are the distinct observed profiles, $N_j = \#\{i : X_i = x^{(j)}\}$, and $N_{1j}(Z) = \sum_{i: X_i = x^{(j)}} Z_i$, then

$$\Delta_Z\{g(X)\} = \sum_{j=1}^J \left\{ \frac{N_{1j}(Z)}{N_1} - \frac{N_j - N_{1j}(Z)}{N_0} \right\} g(x^{(j)}),$$

which is a function of the treatment counts defining $\mathcal{F}_X$.

Recall that $B_X = \mathbb{E}_Z\{\Delta_Z(M) \mid \mathcal{F}_X\}$. We may therefore write

$$\Delta_Z(M) - \Delta_Z\{g(X)\} = \{\Delta_Z(M) - B_X\} + \{B_X - \Delta_Z\{g(X)\}\}.\tag{40}$$

The two terms on the right-hand side are orthogonal in L^2 . Indeed,

$$\begin{aligned}\mathbb{E}_Z [\{\Delta_Z(M) - B_X\} \{B_X - \Delta_Z\{g(X)\}\}] \\ &= \mathbb{E}_Z [\{B_X - \Delta_Z\{g(X)\}\} \mathbb{E}_Z\{\Delta_Z(M) - B_X \mid \mathcal{F}_X\}] \\ &= 0,\end{aligned}$$

because $\mathbb{E}_Z\{\Delta_Z(M) - B_X \mid \mathcal{F}_X\} = \mathbb{E}_Z\{\Delta_Z(M) \mid \mathcal{F}_X\} - B_X = 0$. Squaring (40), taking expectations, and using orthogonality yields

$$\begin{aligned}\mathbb{E}_Z [\{\Delta_Z(M) - \Delta_Z\{g(X)\}\}^2] \\ &= \mathbb{E}_Z [\{\Delta_Z(M) - B_X\}^2] + \mathbb{E}_Z [\{B_X - \Delta_Z\{g(X)\}\}^2].\end{aligned}\tag{41}$$

The first term on the right-hand side of (41) does not depend on g . Consequently,

$$\begin{aligned} \arg \min_{g \in \mathcal{G}} \mathbb{E}_Z [\{\Delta_Z(M) - \Delta_Z\{g(X)\}\}^2] \\ = \arg \min_{g \in \mathcal{G}} \mathbb{E}_Z [\{B_X - \Delta_Z\{g(X)\}\}^2]. \end{aligned} \quad (42)$$

For any fixed $g \in \mathcal{G}$ and $c \in \mathbb{R}$,

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^N \{M_i - c - g_i\}^2 &= \frac{1}{N} \sum_{i=1}^N [R_i(g) - \bar{R}(g) + \bar{R}(g) - c]^2 \\ &= \frac{1}{N} \sum_{i=1}^N \{R_i(g) - \bar{R}(g)\}^2 + \{\bar{R}(g) - c\}^2 \\ &= \frac{N-1}{N} S_{M-g}^2 + \{\bar{M} - \bar{g} - c\}^2, \end{aligned}$$

where the cross-product vanishes because $\sum_i \{R_i(g) - \bar{R}(g)\} = 0$. The expression is uniquely minimized over c at $c_g^* = \bar{R}(g) = \bar{M} - \bar{g}$. Therefore,

$$\min_{c \in \mathbb{R}} \frac{1}{N} \sum_{i=1}^N \{M_i - c - g(X_i)\}^2 = \frac{N-1}{N} S_{M-g}^2. \quad (43)$$

Because $(N-1)/N > 0$ does not depend on g ,

$$\arg \min_{g \in \mathcal{G}} \min_{c \in \mathbb{R}} \frac{1}{N} \sum_{i=1}^N \{M_i - c - g(X_i)\}^2 = \arg \min_{g \in \mathcal{G}} S_{M-g}^2.$$

Combining (39), (42), and (43) establishes

$$\begin{aligned} \arg \min_{g \in \mathcal{G}} \min_{c \in \mathbb{R}} \frac{1}{N} \sum_{i=1}^N \{M_i - c - g(X_i)\}^2 \\ &= \arg \min_{g \in \mathcal{G}} \mathbb{E}_Z [\{\Delta_Z(M) - \Delta_Z\{g(X)\}\}^2] \\ &= \arg \min_{g \in \mathcal{G}} \mathbb{E}_Z [\{B_X - \Delta_Z\{g(X)\}\}^2] \\ &= \arg \min_{g \in \mathcal{G}} \text{MSE}_Z \{\hat{\tau}(g)\} \\ &= \arg \min_{g \in \mathcal{G}} S_{M-g}^2. \end{aligned}$$

Finally, if $\mathcal{G}$ is closed under the addition of constants, the optimal intercept c_g^* can be absorbed

into the score by replacing g with $g + c_g^*$. This replacement does not change the adjustment because

$$\Delta_Z\{g(X) + c_g^*\} = \Delta_Z\{g(X)\}.$$

□

A.5 Proof of Proposition 5

Proof.

For each unit i , recall that $M_i = qY_i(1) + pY_i(0)$. Expanding the design-weighted squared-error loss gives

$$\begin{aligned} & q\{Y_i(1) - g_i\}^2 + p\{Y_i(0) - g_i\}^2 \\ &= qY_i(1)^2 + pY_i(0)^2 - 2g_i\{qY_i(1) + pY_i(0)\} + g_i^2 \\ &= \{M_i - g_i\}^2 + [qY_i(1)^2 + pY_i(0)^2 - M_i^2]. \end{aligned}$$

The remaining term satisfies the weighted-variance identity

$$\begin{aligned} & qY_i(1)^2 + pY_i(0)^2 - \{qY_i(1) + pY_i(0)\}^2 \\ &= pq\{Y_i(1) - Y_i(0)\}^2. \end{aligned}$$

Consequently, unit by unit,

$$q\{Y_i(1) - g_i\}^2 + p\{Y_i(0) - g_i\}^2 = \{M_i - g_i\}^2 + pq\{Y_i(1) - Y_i(0)\}^2. \quad (44)$$

Averaging (44) over the finite population yields

$$\begin{aligned} \mathcal{L}_p(g) &= q\frac{1}{N} \sum_{i=1}^N \{Y_i(1) - g_i\}^2 + p\frac{1}{N} \sum_{i=1}^N \{Y_i(0) - g_i\}^2 \\ &= \frac{1}{N} \sum_{i=1}^N \{M_i - g_i\}^2 + pq\frac{1}{N} \sum_{i=1}^N \{Y_i(1) - Y_i(0)\}^2. \end{aligned}$$

This proves Equation (??). Because the second term does not depend on g ,

$$\arg \min_{g \in \mathcal{G}} \mathcal{L}_p(g) = \arg \min_{g \in \mathcal{G}} \frac{1}{N} \sum_{i=1}^N \{M_i - g(X_i)\}^2.$$

Fix a candidate function g with respect to the treatment assignment. Since $Z_i Y_i = Z_i Y_i(1)$ and

$(1 - Z_i)Y_i = (1 - Z_i)Y_i(0)$, the observable loss can be written as

$$\begin{aligned}\widehat{\mathcal{L}}_p(g) &= q \frac{1}{N_1} \sum_{i=1}^N Z_i \{Y_i(1) - g_i\}^2 \\ &\quad + p \frac{1}{N_0} \sum_{i=1}^N (1 - Z_i) \{Y_i(0) - g_i\}^2.\end{aligned}$$

Under complete randomization, $\mathbb{E}_Z(Z_i) = p = \frac{N_1}{N}$, and $\mathbb{E}_Z(1 - Z_i) = q = \frac{N_0}{N}$. Therefore,

$$\begin{aligned}\mathbb{E}_Z\{\widehat{\mathcal{L}}_p(g)\} &= q \frac{1}{N_1} \sum_{i=1}^N \mathbb{E}_Z(Z_i) \{Y_i(1) - g_i\}^2 \\ &\quad + p \frac{1}{N_0} \sum_{i=1}^N \mathbb{E}_Z(1 - Z_i) \{Y_i(0) - g_i\}^2 \\ &= q \frac{p}{N_1} \sum_{i=1}^N \{Y_i(1) - g_i\}^2 + p \frac{q}{N_0} \sum_{i=1}^N \{Y_i(0) - g_i\}^2 \\ &= q \frac{1}{N} \sum_{i=1}^N \{Y_i(1) - g_i\}^2 + p \frac{1}{N} \sum_{i=1}^N \{Y_i(0) - g_i\}^2 \\ &= \mathcal{L}_p(g),\end{aligned}$$

where the penultimate equality uses $p/N_1 = 1/N$ and $q/N_0 = 1/N$. This establishes the final statement.

The calculation is pointwise in g : it requires g to be fixed with respect to the target treatment assignment. It does not, by itself, imply that $\mathbb{E}_Z\{\widehat{\mathcal{L}}_p(\widehat{g})\} = \mathcal{L}_p(\widehat{g})$ when $\widehat{g}$ is trained adaptively on the same assignment. $\square$

B Consistency and variance estimation for the direct estimator

From semiparametric theory, [Zhang and Ma \(2019, Sections 2.2–2.3 and Appendix\)](#) develop direct learning of an augmentation function, an asymptotic linearization for an estimated augmentation, and variance estimation under an i.i.d. framework. For the mean effect, their optimal function is the centered complementary-weighted score divided by $p(1 - p)$. Their empirical loss uses an estimated, arm-centered influence-function response, whereas our learner uses the raw-outcome complementary-weighted loss. These criteria have the same population target after translation and scaling, but their finite-sample minimizers need not coincide. We complement these results with a finite-population theorem for the specific estimator used here.

Consider a sequence of fixed finite populations indexed by N . Partition the units into a fixed number $V \geq 2$ of folds $I_1, \dots, I_V$, independently of treatment assignment, with prespecified sizes $n_v = |I_v|$. For $i \in I_v$, let $\hat{g}_i = \hat{g}^{(-v)}(X_i)$, where the entire fitting procedure excludes the assignments and outcomes in I_v . The estimator is:

$$\hat{\tau}_{\text{dir}} = \frac{1}{N_1} \sum_{i:Z_i=1} (Y_i - \hat{g}_i) - \frac{1}{N_0} \sum_{i:Z_i=0} (Y_i - \hat{g}_i). \quad (45)$$

Let g_N^* be a reference score deterministic given the finite population, and abbreviate $g_{Ni}^* = g_N^*(X_i)$. The natural choice is the finite-population minimizer of the complementary weighted loss in the prespecified prediction class, although validity does not require that this class correctly specifies the potential outcomes. Define

$$u_i(z) = Y_i(z) - g_{Ni}^*, \quad r_i = q_N u_i(1) + p_N u_i(0) = M_i - g_{Ni}^*, \quad M_i = q_N Y_i(1) + p_N Y_i(0). \quad (46)$$

All probability statements below condition on the finite population. When the split or the learning algorithm is randomized, probability also averages over its random seed, which is independent of treatment assignment.

Assumption 1 (Conditions for direct-score inference). *As $N \rightarrow \infty$, the following conditions hold.*

1. *Treatment is completely randomized with N_1 treated units, $p_N = \frac{N_1}{N} \rightarrow p_\infty \in (0, 1)$, and $n_v/N \rightarrow \rho_v \in (0, 1)$ for every v . The fold-specific fitting procedure is restricted to training information as described above.*
2. *The fitted scores converge in held-out empirical squared error to a common fixed reference score:*

$$Q_N := \max_{1 \leq v \leq V} \frac{1}{n_v} \sum_{i \in I_v} \{\hat{g}^{(-v)}(X_i) - g_{Ni}^*\}^2 = o_{\mathbb{P}}(1). \quad (47)$$

3. *For $z \in \{0, 1\}$,*

$$\begin{aligned} \sup_N \frac{1}{N} \sum_{i=1}^N \{u_i(z) - \bar{u}(z)\}^4 &< \infty, \\ \sigma_N^2 := \frac{S_r^2}{p_N q_N} &\longrightarrow \sigma^2 \in (0, \infty), \quad S_\tau^2 \longrightarrow s_\tau^2 < \infty. \end{aligned} \quad (48)$$

Condition (47) is a prediction-stability condition, not a requirement that the unobserved M_i be recovered without residual error. It requires convergence to g_N^* , not convergence to M . It permits misspecification and imposes no $N^{-1/4}$ convergence rate. Appendix ?? gives sufficient condi-

tions for weighted empirical risk minimization and direct weighted LASSO. The fourth-moment condition is a convenient sufficient condition for the finite-population central limit theorem and consistency of residual sample variances; it can be weakened.

For estimation of uncertainty, let

$$\hat{u}_i = Y_i - \hat{g}_i, \quad \bar{\hat{u}}_z = \frac{1}{N_z} \sum_{i:Z_i=z} \hat{u}_i, \quad \hat{s}_z^2 = \frac{1}{N_z - 1} \sum_{i:Z_i=z} (\hat{u}_i - \bar{\hat{u}}_z)^2, \quad (49)$$

and define $\hat{V}_{\text{dir}} = \frac{\hat{s}_1^2}{N_1} + \frac{\hat{s}_0^2}{N_0}$.

Theorem 1 (Consistency, asymptotic normality, and conservative inference). *Under Assumption 1,*

$$\hat{\tau}_{\text{dir}} - \tau_N = \Delta_Z(r) + o_{\mathbb{P}}(N^{-1/2}), \quad \sqrt{N}(\hat{\tau}_{\text{dir}} - \tau_N) \xrightarrow{d} \mathcal{N}(0, \sigma^2). \quad (50)$$

The exact variance of the fixed-score leading term is

$$V_N^* := \text{Var}_Z\{\Delta_Z(r)\} = \frac{S_r^2}{Np_Nq_N} = \frac{S_{u(1)}^2}{N_1} + \frac{S_{u(0)}^2}{N_0} - \frac{S_\tau^2}{N}. \quad (51)$$

and,

$$\left[\hat{\tau}_{\text{dir}} - z_{1-\alpha/2} \sqrt{\hat{V}_{\text{dir}}}, \quad \hat{\tau}_{\text{dir}} + z_{1-\alpha/2} \sqrt{\hat{V}_{\text{dir}}} \right] \quad (52)$$

has limiting randomization coverage at least $1 - \alpha$, where z_a is the a th standard normal quantile.

Its limiting coverage is exactly $1 - \alpha$ when $s_\tau^2 = 0$.

The distinction between V_N^* and $\hat{V}_{\text{dir}}$ is important. Because $u_i(1) - u_i(0) = \tau_i$, the unidentifiable variance of individual treatment effects remains in the exact finite-population variance. Without additional restrictions, the proposed estimator consistently estimates the Neyman upper bound at the N^{-1} scale, not necessarily the exact variance at that scale.

Write $p = p_N$ and $q = q_N$ when there is no ambiguity. The proof needs some lemmas.

Lemma 1 (Exact identities). *For any fixed vector A ,*

$$\Delta_Z(A) = \frac{1}{Npq} \sum_{i=1}^N (Z_i - p) A_i, \quad \text{Var}_Z\{\Delta_Z(A)\} = \frac{S_A^2}{Npq}. \quad (53)$$

For the fixed reference score in (46),

$$\hat{\tau}(g_N^*) - \tau_N = \Delta_Z(r), \quad \frac{S_r^2}{Npq} = \frac{S_{u(1)}^2}{N_1} + \frac{S_{u(0)}^2}{N_0} - \frac{S_\tau^2}{N}. \quad (54)$$

For any fitted numerical score vector, whether or not it is cross-fitted, put $e_i = \hat{g}_i - g_{Ni}^$. Then the*

following algebraic identity holds for every realized assignment:

$$\hat{\tau}_{\text{dir}} - \tau_N = \Delta_Z(r) - \Delta_Z(e). \quad (55)$$

Proof. The first identity follows from $N_1 = Np$ and $N_0 = Nq$. Under complete randomization,

$$\mathbb{E}_Z Z_i = p, \quad \text{Var}_Z(Z_i) = pq, \quad \text{Cov}_Z(Z_i, Z_j) = -\frac{pq}{N-1} \quad (i \neq j).$$

Consequently,

$$\begin{aligned} \text{Var}_Z \left\{ \sum_i (Z_i - p) A_i \right\} &= pq \sum_i A_i^2 - \frac{pq}{N-1} \sum_{i \neq j} A_i A_j \\ &= Npq S_A^2. \end{aligned}$$

Dividing by $(Npq)^2$ proves the variance identity. Our main text shows $\hat{\tau}_{\text{DM}} - \tau_N = \Delta_Z(M)$ gives $\hat{\tau}(g_N^*) - \tau_N = \Delta_Z(M - g_N^*) = \Delta_Z(r)$. Let $S_{u(1),u(0)}$ denote the finite-population covariance. Because $r = qu(1) + pu(0)$ and $\tau = u(1) - u(0)$,

$$\begin{aligned} \frac{S_r^2}{pq} &= \frac{q}{p} S_{u(1)}^2 + \frac{p}{q} S_{u(0)}^2 + 2S_{u(1),u(0)} \\ &= \frac{S_{u(1)}^2}{p} + \frac{S_{u(0)}^2}{q} - \{S_{u(1)}^2 + S_{u(0)}^2 - 2S_{u(1),u(0)}\} \\ &= \frac{S_{u(1)}^2}{p} + \frac{S_{u(0)}^2}{q} - S_\tau^2. \end{aligned}$$

Finally, subtracting $\Delta_Z(\hat{g})$ instead of $\Delta_Z(g_N^*)$ yields (55). This last step is algebra and therefore does not require independence of $\hat{g}$ and Z . $\square$

Proposition B.1. Suppose $p_N \rightarrow p_\infty \in (0, 1)$, $S_r^2 = O(1)$, and $N^{-1} \sum_i e_i^2 = o_{\mathbb{P}}(1)$. Then $\hat{\tau}_{\text{dir}} - \tau_N = o_{\mathbb{P}}(1)$.

Proof. Lemma 1 and Chebyshev's inequality imply $\Delta_Z(r) = O_{\mathbb{P}}(N^{-1/2})$. Since $\sum_i (Z_i - p)^2 = Npq$ under complete randomization, Cauchy–Schwarz gives

$$|\Delta_Z(e)|^2 \leq \frac{1}{pq} \frac{1}{N} \sum_i e_i^2 = o_{\mathbb{P}}(1). \quad (56)$$

Apply (55). $\square$

Lemma 2 (Negligible score-estimation imbalance). *Under conditions 1 and 2 of Assumption 1,*

$$\Delta_Z(e) = o_{\mathbb{P}}(N^{-1/2}). \quad (57)$$

Proof. Fix a fold v . Let $\mathcal{H}_v$ contain the fold partition, all pretreatment covariates, independent algorithmic random seeds, and all assignments and observed outcomes outside I_v . Conditional on $\mathcal{H}_v$, each error $e_{vi} = \widehat{g}^{(-v)}(X_i) - g_{Ni}^*$, $i \in I_v$, is fixed. Furthermore, the number of treated units in the held-out fold is known:

$$m_v = N_1 - \sum_{i \notin I_v} Z_i, \quad p_v = m_v/n_v.$$

The conditional assignment in I_v is uniform over subsets of size m_v . Writing $\bar{e}_v = n_v^{-1} \sum_{i \in I_v} e_{vi}$, decompose

$$\sum_{i \in I_v} (Z_i - p) e_{vi} = \underbrace{\sum_{i \in I_v} (Z_i - p_v) e_{vi}}_{A_v} + \underbrace{(m_v - pn_v) \bar{e}_v}_{B_v}. \quad (58)$$

The first term has conditional mean zero and conditional variance

$$\mathbb{E}(A_v \mid \mathcal{H}_v) = 0, \quad \text{Var}(A_v \mid \mathcal{H}_v) = n_v p_v (1 - p_v) S_{e,v}^2,$$

where $S_{e,v}^2 = (n_v - 1)^{-1} \sum_{i \in I_v} (e_{vi} - \bar{e}_v)^2$. This formula also gives zero when $m_v \in \{0, n_v\}$. Put $Q_v = n_v^{-1} \sum_{i \in I_v} e_{vi}^2$. For all sufficiently large N , $S_{e,v}^2 \leq 2Q_v$, and conditional Chebyshev's inequality implies

$$\mathbb{P}\left(|A_v| > \epsilon \sqrt{N} \mid \mathcal{H}_v\right) \leq \min\{1, CQ_v/\epsilon^2\}.$$

Here and below C denotes a finite constant independent of N . Since $Q_v = o_{\mathbb{P}}(1)$ and the right side is bounded, its expectation tends to zero. Hence $A_v/\sqrt{N} = o_{\mathbb{P}}(1)$.

Conditional on the assignment-independent partition, m_v is hypergeometric, with

$$\mathbb{E}(m_v) = pn_v, \text{ and } \text{Var}(m_v) = n_v pq \frac{N - n_v}{N - 1} = O(N).$$

Therefore $(m_v - pn_v)/\sqrt{N} = O_{\mathbb{P}}(1)$. Also, $|\bar{e}_v| \leq \sqrt{Q_v} = o_{\mathbb{P}}(1)$, so $B_v/\sqrt{N} = o_{\mathbb{P}}(1)$. This product argument does not require independence between the fold count and the fitted score. Summing

over the fixed number of folds gives

$$\sqrt{N} \Delta_Z(e) = \frac{1}{pq} \sum_{v=1}^V \frac{A_v + B_v}{\sqrt{N}} = o_{\mathbb{P}}(1).$$

□

Proof of Theorem 1

Proof. Combining (55) and Lemma 2 proves the first assertion in (50). In view of $\text{Var}_Z\{\Delta_Z(r)\} = S_r^2/(Npq) = O(N^{-1})$, it also proves consistency.

For Finite-population central limit theorem, set $\tilde{u}_i(z) = u_i(z) - \bar{u}(z)$ and $\tilde{r}_i = r_i - \bar{r} = q\tilde{u}_i(1) + p\tilde{u}_i(0)$. Convexity of $x \mapsto x^4$ and the moment assumption imply

$$\frac{1}{N} \sum_i \tilde{r}_i^4 \leq q \frac{1}{N} \sum_i \tilde{u}_i(1)^4 + p \frac{1}{N} \sum_i \tilde{u}_i(0)^4 = O(1).$$

Consequently,

$$\max_i \tilde{r}_i^2 \leq \left(\sum_i \tilde{r}_i^4 \right)^{1/2} = O(N^{1/2}), \text{ and } \frac{\max_i \tilde{r}_i^2}{\min(N_1, N_0) S_r^2} \longrightarrow 0,$$

since $S_r^2/(pq) \rightarrow \sigma^2 > 0$ and p, q are bounded away from zero. The treated units form a simple random sample without replacement from the finite population. The finite-population central limit theorem (Li and Ding, 2017, Theorem 1) therefore applies to its mean of r . Because $\Delta_Z(r) = (\bar{r}_1 - \bar{r})/q$,

$$\frac{\Delta_Z(r)}{\sqrt{S_r^2/(Npq)}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Slutsky's theorem and Step 1 prove the normal limit in (50). Equation (51) follows from Lemma 1.

Now, we show variance estimation. For $z \in \{0, 1\}$, let $s_{z,*}^2$ be the sample variance, among units with $Z_i = z$, of the fixed values $u_i(z)$. Define $a_i = u_i(z) - \bar{u}(z)$. The arm is a simple random sample of size N_z , so its sample mean of a_i is $O_{\mathbb{P}}(N^{-1/2})$. The fourth-moment assumption gives

$$S_{a^2}^2 = \frac{1}{N-1} \sum_i \left(a_i^2 - \frac{1}{N} \sum_j a_j^2 \right)^2 \leq \frac{1}{N-1} \sum_i a_i^4 = O(1).$$

The simple-random-sample variance formula applied to a_i^2 then yields

$$\frac{1}{N_z} \sum_{i:Z_i=z} a_i^2 - \frac{1}{N} \sum_i a_i^2 = o_{\mathbb{P}}(1).$$

Using

$$s_{z,*}^2 = \frac{N_z}{N_z - 1} \left\{ \frac{1}{N_z} \sum_{i:Z_i=z} a_i^2 - \left(\frac{1}{N_z} \sum_{i:Z_i=z} a_i \right)^2 \right\},$$

we conclude that $s_{z,*}^2 - S_{u(z)}^2 = o_{\mathbb{P}}(1)$.

Let $s_{e,z}^2$ be the sample variance of e_i in arm z . Since $N^{-1} \sum_i e_i^2 \leq Q_N = o_{\mathbb{P}}(1)$,

$$s_{e,z}^2 \leq \frac{1}{N_z - 1} \sum_{i:Z_i=z} e_i^2 \leq \frac{1}{N_z - 1} \sum_i e_i^2 = o_{\mathbb{P}}(1).$$

Within arm z , $\hat{u}_i = u_i(z) - e_i$. The sample covariance inequality therefore gives

$$|\hat{s}_z^2 - s_{z,*}^2| \leq 2\sqrt{s_{z,*}^2 s_{e,z}^2} + s_{e,z}^2 = o_{\mathbb{P}}(1).$$

No independence between residuals and predictions is required for this inequality. Thus $\hat{s}_z^2 = S_{u(z)}^2 + o_{\mathbb{P}}(1)$, and

$$\begin{aligned} N\hat{V}_{\text{dir}} &= \frac{S_{u(1)}^2}{p} + \frac{S_{u(0)}^2}{q} + o_{\mathbb{P}}(1) \\ &= \frac{S_r^2}{pq} + S_{\tau}^2 + o_{\mathbb{P}}(1). \end{aligned}$$

Therefore, $N(\hat{V}_{\text{dir}} - V_N^*) - S_{\tau}^2 = o_{\mathbb{P}}(1)$, and $N\hat{V}_{\text{dir}} \xrightarrow{\mathbb{P}} \sigma^2 + s_{\tau}^2$.

Another application of Slutsky's theorem gives

$$\frac{\hat{\tau}_{\text{dir}} - \tau_N}{\sqrt{\hat{V}_{\text{dir}}}} \xrightarrow{d} \mathcal{N}\left(0, \frac{\sigma^2}{\sigma^2 + s_{\tau}^2}\right).$$

The limiting coverage of (52) is consequently

$$2\Phi\left(z_{1-\alpha/2}\sqrt{1 + s_{\tau}^2/\sigma^2}\right) - 1 \geq 1 - \alpha.$$

It equals $1 - \alpha$ when $s_{\tau}^2 = 0$. □

Corollary 1 (Attainment of the class-oracle variance). *Suppose the reference score minimizes the*

complementary weighted finite-population loss over a class $\mathcal{G}_N$ that contains the constant scores and is closed under adding constants. If Assumption 1 holds, the direct estimator has the same first-order distribution as the best fixed-score estimator in $\mathcal{G}_N$. In particular,

$$V_N^* \leq \frac{S_M^2}{Np_Nq_N} = \text{Var}_Z(\hat{\tau}_{\text{DM}}). \quad (59)$$

Proof. For every numerical score vector g , completing the square gives

$$\mathcal{L}_p(g) = \frac{1}{N} \sum_i (M_i - g_i)^2 + pq \frac{1}{N} \sum_i \tau_i^2. \quad (60)$$

The second term does not depend on g . Because the class is closed under adding constants, minimizing the first term also chooses an intercept such that $\overline{M - g} = 0$. More explicitly, for every candidate g , $g + \overline{M - g}$ belongs to the class and has squared prediction error $(N - 1)S_{M-g}^2/N$. Hence minimizing $\mathcal{L}_p$ is equivalent to minimizing S_{M-g}^2 over this class. A constant score has $S_{M-g}^2 = S_M^2$, which proves (59). The oracle first-order distribution follows from Theorem 1. $\square$

C Simulation implementation details

Full-sample estimators. The unadjusted estimator is $\hat{\tau}_{\text{DM}} = \bar{Y}_1 - \bar{Y}_0$. All- X OLS reports the coefficient on Z from an additive regression of Y on an intercept, Z , and all 60 full-sample-centered covariates. All- X Lin additionally interacts Z with every centered covariate. If $\hat{\beta}_A$ is the additive slope and $\hat{\beta}_1, \hat{\beta}_0$ are the within-arm slopes, these two estimators can be written as

$$\begin{aligned} \hat{\tau}_{\text{OLS,all}} &= \hat{\tau}_{\text{DM}} - \hat{\beta}_A^\top \Delta_Z(X), \\ \hat{\tau}_{\text{Lin,all}} &= \hat{\tau}_{\text{DM}} - \{q\hat{\beta}_1 + p\hat{\beta}_0\}^\top \Delta_Z(X). \end{aligned} \quad (61)$$

Both estimators are identified with $K = 60$; even under $p = 1/4$, the fully interacted Lin fit has 100 treated observations for 60 covariates and an intercept.

One-step LASSO is computed from the full analysis sample by solving:

$$(\hat{\alpha}_\lambda, \hat{\tau}_{1S}, \hat{\beta}_\lambda) \in \arg \min_{\alpha, t, \beta} \left[\frac{1}{2N} \sum_{i=1}^N \{Y_i - \alpha - tZ_i - (X_i - \bar{X})^\top \beta\}^2 + \lambda \|\beta\|_1 \right]. \quad (62)$$

The intercept and treatment coefficient are unpenalized, whereas all 60 covariate coefficients receive the LASSO penalty. We choose λ by five-fold cross-validation with folds stratified by treatment status, using the minimum-CV-error rule over a 40-value penalty path. The reported estimate

is the full-sample coefficient $\hat{\tau}_{1S}$; there is no outer cross-fitting or post-LASSO OLS refit. The first-order conditions for the two unpenalized coefficients imply the realized-sample identity

$$\hat{\tau}_{1S} = \hat{\tau}_{DM} - \hat{\beta}_\lambda^\top \Delta_Z(X). \quad (63)$$

Unpenalizing Z therefore prevents direct shrinkage of the treatment coefficient, although it does not by itself imply exact finite-sample design unbiasedness for the adaptive estimator.

Direct and Arm-specific estimators. Direct M -LASSO and Arm-specific LASSO/AIPW use the same four-fold outer cross-fitting scheme to estimate $\hat{g}(x)$ and construct final corrected estimator. We form the folds separately within treatment and control so that every fold has treatment fraction p . In each outer training sample, Direct M -LASSO solves

$$\hat{g}_D^{(-v)} \in \arg \min_{\alpha, \beta} \left[\frac{1}{2|I_v^c|} \sum_{i \notin I_v} w_i \{Y_i - \alpha - X_i^\top \beta\}^2 + \lambda \|\beta\|_1 \right], \text{ where } w_i = \begin{cases} q/p, & Z_i = 1, \\ p/q, & Z_i = 0. \end{cases} \quad (64)$$

Using the same folds, Arm-specific LASSO fits $\hat{m}_1^{(-v)}$ and $\hat{m}_0^{(-v)}$ separately and combines their held-out predictions as $\hat{g}_{AS,i} = q\hat{m}_1^{(-v)}(X_i) + p\hat{m}_0^{(-v)}(X_i)$, $i \in I_v$. Within each outer training sample, every LASSO is tuned by three-fold inner cross-validation using the minimum-CV-error rule over 40 penalties. All preprocessing, fitting, and tuning are confined to the training observations. For all three LASSO procedures, the implementation standardizes covariates within the observations supplied to each fit and returns coefficients on the original scale. For notational simplicity, the displayed objectives suppress this internal rescaling.

For either held-out score $\hat{g}$, the associated estimator is

$$\hat{\tau}_{CF}(\hat{g}) = \sum_{v=1}^4 \frac{|I_v|}{N} \left\{ \overline{Y - \hat{g}_{1,v}} - \overline{Y - \hat{g}_{0,v}} \right\}. \quad (65)$$

For the arm-specific method, this expression is algebraically identical to the cross-fitted augmented inverse-probability-weighted estimator with the known randomization probability:

$$\begin{aligned} \hat{\tau}_{AS} &= \frac{1}{N} \sum_{i=1}^N \left[\hat{m}_{1i} - \hat{m}_{0i} + \frac{Z_i}{p} \{Y_i - \hat{m}_{1i}\} - \frac{1 - Z_i}{q} \{Y_i - \hat{m}_{0i}\} \right] \\ &= \hat{\tau}_{CF}(\hat{g}_{AS}). \end{aligned} \quad (66)$$

Thus, the arm-specific learner supplies the nuisance outcome regressions for the familiar known-propensity AIPW form (Robins et al., 1994, 1995); cross-fitting changes how the nuisance func-

tions are learned, not the algebraic equivalence. This is also the generalized-regression representation in Equation (23).

Oracle and identity check. For the DGP oracle, we compute

$$g_s^*(X) = \mathbb{E}(M \mid X) = \sqrt{R_M^2} g_s^\circ(X), \quad \hat{\tau}_{\text{or}} = \hat{\tau}_{\text{DM}} - \Delta_Z\{g_s^*(X)\}. \quad (67)$$

This adjustment removes the covariate-predictable component of M but leaves the irreducible component. Adjustment by the full unit-level M is used only as an internal identity check.

D Covariate Selection Using an Independent Sample

To avoid additional inferential complications from selecting covariates in the target experiment, we assume that an independent sample is available for covariate selection and external-score fitting. This supplementary study asks which criterion should determine whether a covariate enters adjustment. In particular, it examines whether selecting predictors of the control outcome is sufficient for selecting predictors of ATE estimation error. The design is deliberately one-sided. It contains a setting in which control-outcome prognosticity does not translate into useful error prediction, but it does not contain the converse case.

D.1 Design and covariate-set rules

Each replication contains independently generated selection and target samples with

$$N_{\text{sel}} = N_{\text{target}} = 400, K = 20. \quad (68)$$

The two populations use the same allocation fraction and the same structural scenario from Section 5.2. The first six covariates retain the roles specified there; the remaining covariates are irrelevant. Because normalization is fixed by Σ , coefficients fitted in the selection sample and covariates in the target sample are expressed in the same units. Outcome-dependent selection and external-score fitting use only outcomes from the selection sample. The raw-balance rule uses target assignments and covariates but never target outcomes.

We compare five covariate sets. The control-outcome set $\hat{\mathcal{J}}_0$ is the support selected by a LASSO of Y on X among controls in the selection sample. The raw-imbalance set is

$$\hat{\mathcal{J}}_{\text{imb}}(Z_T) = \{j : p_{\text{bal},j}(Z_T) < 0.10\}, \quad (69)$$

where $p_{\text{bal},j}$ is the two-sided standardized-mean-difference test for target covariate j . We also use the intersection and union

$$\hat{\mathcal{J}}_{\cap} = \hat{\mathcal{J}}_0 \cap \hat{\mathcal{J}}_{\text{imb}}, \quad \hat{\mathcal{J}}_{\cup} = \hat{\mathcal{J}}_0 \cup \hat{\mathcal{J}}_{\text{imb}}. \quad (70)$$

Finally, the design-relevant set $\hat{\mathcal{J}}_M$ is selected from both treatment arms in the selection sample by the complementary weighted LASSO loss

$$\frac{1}{N_{\text{sel}}} \sum_{i=1}^{N_{\text{sel}}} w_i^S \{Y_i^S - \alpha - (X_i^S)^\top \beta\}^2 + \lambda_M \|\beta\|_1, \quad w_i^S = \begin{cases} q/p, & Z_i^S = 1, \\ p/q, & Z_i^S = 0. \end{cases} \quad (71)$$

Both LASSO selection rules use five-fold cross-validation and the one-standard-error rule. Folds for $\hat{\mathcal{J}}_0$ are formed within controls in the selection sample, whereas folds for $\hat{\mathcal{J}}_M$ are stratified by treatment status in that sample.

D.2 Common post-selection estimators

To isolate the effect of selecting different sets, every nonempty set $\mathcal{J}$ receives the same unpenalized complementary-weighted refit in the selection sample. If $\hat{\gamma}_{\mathcal{J}}$ denotes the resulting coefficient, the primary estimator applies the external score to the target experiment:

$$\hat{\tau}_{\text{ext}}(\mathcal{J}) = \hat{\tau}_{\text{DM}}^T - \hat{\gamma}_{\mathcal{J}}^\top \Delta_{Z_T}(X_{\mathcal{J}}^T). \quad (72)$$

An empty set produces no adjustment. As a secondary practical comparison, we apply target-sample Lin regression to the same selected set:

$$\hat{\tau}_{\text{Lin}}(\mathcal{J}) = \hat{\tau}_{\text{DM}}^T - \{q\hat{\beta}_{1,\mathcal{J}}^T + p\hat{\beta}_{0,\mathcal{J}}^T\}^\top \Delta_{Z_T}(X_{\mathcal{J}}^T). \quad (73)$$

For the external score, we also report centered prediction risk in the independently generated target population:

$$\text{RelPredRisk}_{M,r}(\hat{g}) = \frac{\sum_{i=1}^N \left[(M_{ir} - \bar{M}_r) - \{\hat{g}_r(X_{ir}) - \bar{g}_r\} \right]^2}{\sum_{i=1}^N (M_{ir} - \bar{M}_r)^2}. \quad (74)$$

Because score fitting occurs in an independent selection sample, this is genuine out-of-sample prediction risk. The mean number of selected covariates is a secondary diagnostic.

D.3 Results

Table A.1 reports the complete comparison, and Figure A.1 displays relative MSE. When control-outcome and design-relevant prognosticity coincide, the two selection rules perform nearly identically. In the sparse-signal scenario, both attain external-score relative MSE 0.506, close to the DGP-oracle value 0.498.

Table A.1: Separate-sample covariate selection and ATE performance

Scenario	Rule	External rel. MSE (MCSE)	Lin rel. MSE (MCSE)	M risk	Size
Null	No adjustment	1.000 (0.000)	1.000 (0.000)	1.000	0.00
Null	Control-outcome	1.000 (0.000)	1.000 (0.000)	1.000	0.01
Null	Raw imbalance	1.020 (0.003)	1.022 (0.003)	1.005	2.00
Null	Intersection	1.000 (0.000)	1.000 (0.000)	1.000	0.00
Null	Union	1.020 (0.003)	1.022 (0.003)	1.005	2.01
Null	Design-relevant	1.000 (0.000)	1.000 (0.000)	1.000	0.00
Sparse M signal	No adjustment	1.000 (0.000)	1.000 (0.000)	1.000	0.00
Sparse M signal	Control-outcome	0.506 (0.008)	0.504 (0.008)	0.508	4.34
Sparse M signal	Raw imbalance	0.749 (0.010)	0.748 (0.010)	0.931	2.01
Sparse M signal	Intersection	0.737 (0.010)	0.734 (0.010)	0.928	0.43
Sparse M signal	Union	0.515 (0.008)	0.515 (0.008)	0.510	5.92
Sparse M signal	Design-relevant	0.506 (0.008)	0.504 (0.008)	0.507	4.15
Arm-specific cancellation	No adjustment	1.000 (0.000)	1.000 (0.000)	1.000	0.00
Arm-specific cancellation	Control-outcome	1.022 (0.004)	1.010 (0.002)	1.026	3.55
Arm-specific cancellation	Raw imbalance	1.044 (0.005)	1.047 (0.005)	1.010	2.01
Arm-specific cancellation	Intersection	1.010 (0.002)	1.007 (0.002)	1.002	0.36
Arm-specific cancellation	Union	1.056 (0.006)	1.034 (0.004)	1.033	5.20
Arm-specific cancellation	Design-relevant	1.000 (0.000)	1.000 (0.000)	1.000	0.00
Unequal allocation with HTE	No adjustment	1.000 (0.000)	1.000 (0.000)	1.000	0.00
Unequal allocation with HTE	Control-outcome	0.545 (0.009)	0.542 (0.009)	0.528	7.29
Unequal allocation with HTE	Raw imbalance	0.796 (0.011)	0.790 (0.011)	0.941	2.01
Unequal allocation with HTE	Intersection	0.758 (0.010)	0.755 (0.010)	0.933	0.73
Unequal allocation with HTE	Union	0.565 (0.009)	0.560 (0.009)	0.533	8.57
Unequal allocation with HTE	Design-relevant	0.535 (0.009)	0.530 (0.009)	0.518	3.29

Notes: Relative MSE is calculated against no adjustment within each scenario; parentheses contain paired Monte Carlo standard errors. The M -risk column reports Equation (74), and Size is the mean number of selected covariates. The DGP oracle’s relative MSE is 1.000, 0.498, 1.000, and 0.516 in the four scenarios, respectively.

Arm-specific cancellation most clearly distinguishes the two prediction targets. The control-outcome rule selects 3.55 covariates on average and includes each heterogeneity-only covariate with probability above 0.999. Although these covariates predict $Y(0)$, they do not predict M . The fitted score therefore has centered prediction risk 1.026 and raises relative MSE to 1.022. The design-relevant rule selects essentially no covariates, has prediction risk 1.000, and reproduces no adjustment. This establishes the intended one-way conclusion: control-outcome prognosticity is not sufficient for prognosticity with respect to ATE estimation error.

Under unequal allocation, the same distinction is smaller but remains visible. The control-outcome rule selects the three heterogeneity-only variables along with the signal variables, yielding

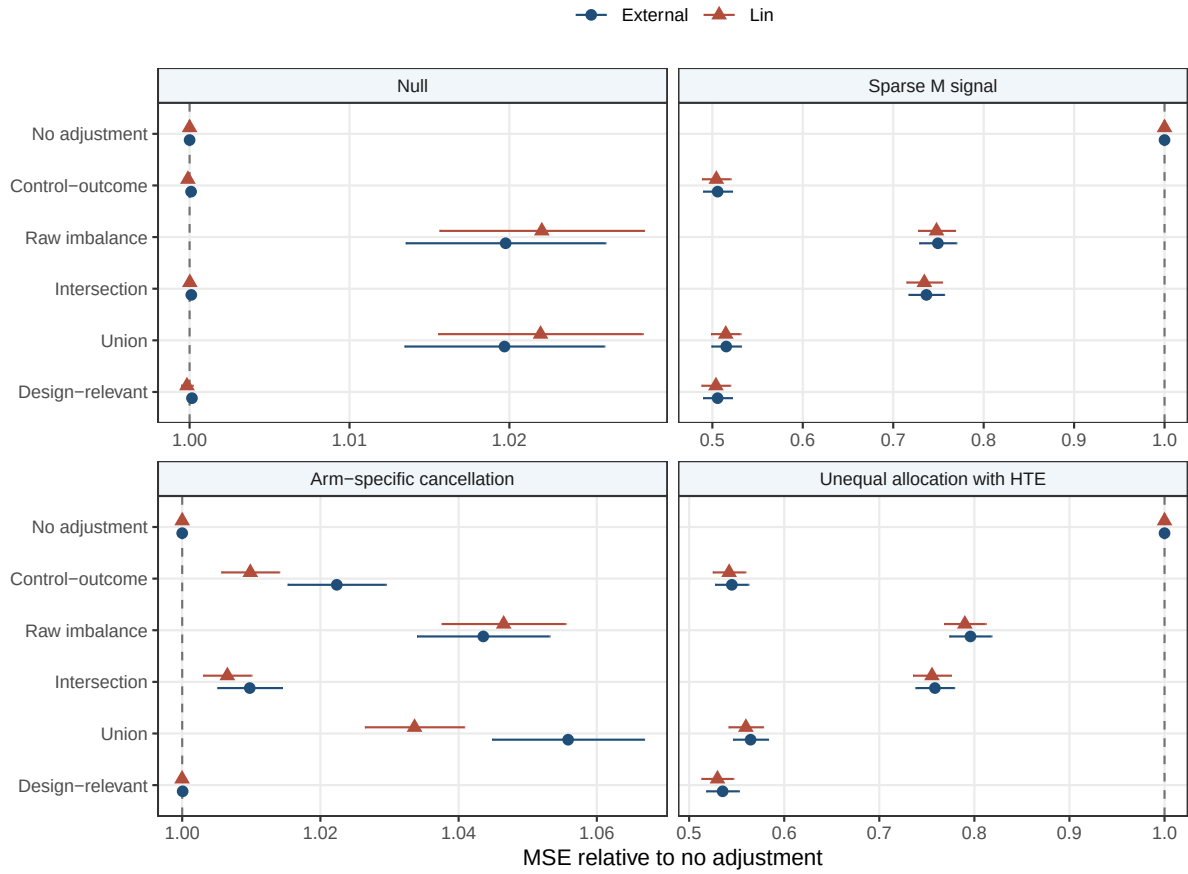


Figure A.1: Separate-sample covariate selection: MSE relative to no adjustment
Notes: Points show relative MSE, and horizontal intervals equal the estimate plus or minus 1.96 paired Monte Carlo standard errors. The dashed vertical line marks the MSE of the unadjusted difference in means.

a mean set size of 7.29 and relative MSE 0.545. The design-relevant rule largely excludes the heterogeneity-only variables, selects 3.29 covariates on average, and lowers relative MSE to 0.535, close to the oracle value 0.516.

The balance-based rules are less reliable. Raw-imbalance selection chooses about two variables even under the null and raises relative MSE to 1.020. In the sparse-signal and unequal-allocation scenarios, it captures too little useful signal, yielding relative MSE 0.749 and 0.796. The union preserves the control-outcome predictors but adds unnecessary variables, producing relative MSE 0.515 under sparse signal, 1.056 under cancellation, and 0.565 under unequal allocation. The target-sample Lin panel preserves the same qualitative ranking, indicating that the main contrast concerns what is selected rather than where the final coefficients are estimated.

The study uses 8,000 independent selection–target sample pairs per scenario. We retain the common replication count used in the archived run; four raw-imbalance or intersection comparisons have paired MCSEs slightly above 0.01, and the largest is 0.0113.

E Small-Sample Estimation-Cost Results

Table A.2 supplements Figure 2 by reporting the complete results for all seven estimators. The design holds $K = 60$ and $p = 1/2$ fixed and varies $N \in \{160, 240, 400\}$ in the sparse-signal and arm-specific-cancellation scenarios. The $N = 400$ entries are the corresponding results from the baseline simulation.

Table A.2: Small-sample estimation cost relative to the DGP oracle

Scenario	N	Method	Bias (MCSE)	RMSE	Rel. MSE (MCSE)	Oracle gap (MCSE) [95% MCI]
Sparse M signal	160	DM	0.0107 (0.0051)	0.1612	1.000 (0.000)	0.513 (0.022) [0.470, 0.556]
Sparse M signal	160	All-X OLS	0.0132 (0.0046)	0.1471	0.832 (0.044)	0.346 (0.034) [0.280, 0.412]
Sparse M signal	160	All-X Lin	0.0143 (0.0059)	0.1874	1.351 (0.083)	0.865 (0.075) [0.718, 1.011]
Sparse M signal	160	One-step LASSO	0.0105 (0.0038)	0.1194	0.549 (0.020)	0.062 (0.009) [0.045, 0.079]
Sparse M signal	160	Direct M-LASSO	0.0104 (0.0038)	0.1211	0.565 (0.020)	0.078 (0.011) [0.057, 0.099]
Sparse M signal	160	Arm-specific LASSO (AIPW)	0.0123 (0.0040)	0.1261	0.612 (0.018)	0.125 (0.014) [0.098, 0.151]
Sparse M signal	160	DGP oracle	0.0099 (0.0035)	0.1125	0.487 (0.022)	0.000 (0.000) [0.000, 0.000]
Sparse M signal	240	DM	-0.0064 (0.0042)	0.1328	1.000 (0.000)	0.504 (0.022) [0.462, 0.547]
Sparse M signal	240	All-X OLS	-0.0029 (0.0034)	0.1073	0.654 (0.031)	0.158 (0.019) [0.120, 0.195]
Sparse M signal	240	All-X Lin	-0.0047 (0.0036)	0.1143	0.741 (0.038)	0.245 (0.027) [0.192, 0.297]
Sparse M signal	240	One-step LASSO	-0.0029 (0.0030)	0.0964	0.527 (0.019)	0.031 (0.007) [0.017, 0.045]
Sparse M signal	240	Direct M-LASSO	-0.0030 (0.0031)	0.0979	0.543 (0.019)	0.048 (0.008) [0.031, 0.064]
Sparse M signal	240	Arm-specific LASSO (AIPW)	-0.0036 (0.0032)	0.1002	0.569 (0.017)	0.073 (0.010) [0.053, 0.093]
Sparse M signal	240	DGP oracle	-0.0014 (0.0030)	0.0935	0.496 (0.022)	0.000 (0.000) [0.000, 0.000]
Sparse M signal	400	DM	-0.0009 (0.0033)	0.1038	1.000 (0.000)	0.515 (0.022) [0.473, 0.558]
Sparse M signal	400	All-X OLS	-0.0010 (0.0024)	0.0767	0.546 (0.026)	0.061 (0.014) [0.034, 0.088]
Sparse M signal	400	All-X Lin	-0.0008 (0.0025)	0.0779	0.563 (0.027)	0.079 (0.015) [0.049, 0.108]
Sparse M signal	400	One-step LASSO	-0.0010 (0.0023)	0.0736	0.502 (0.020)	0.017 (0.005) [0.007, 0.028]
Sparse M signal	400	Direct M-LASSO	-0.0008 (0.0024)	0.0743	0.512 (0.019)	0.028 (0.006) [0.015, 0.040]
Sparse M signal	400	Arm-specific LASSO (AIPW)	-0.0007 (0.0024)	0.0756	0.531 (0.019)	0.046 (0.008) [0.031, 0.061]
Sparse M signal	400	DGP oracle	-0.0006 (0.0023)	0.0723	0.485 (0.022)	0.000 (0.000) [0.000, 0.000]
Arm-specific cancellation	160	DM	0.0029 (0.0050)	0.1576	1.000 (0.000)	0.000 (0.000) [0.000, 0.000]
Arm-specific cancellation	160	All-X OLS	0.0090 (0.0076)	0.2397	2.312 (0.112)	1.312 (0.112) [1.091, 1.532]
Arm-specific cancellation	160	All-X Lin	0.0036 (0.0080)	0.2518	2.552 (0.136)	1.552 (0.136) [1.286, 1.819]
Arm-specific cancellation	160	One-step LASSO	0.0037 (0.0050)	0.1590	1.017 (0.008)	0.017 (0.008) [0.001, 0.032]
Arm-specific cancellation	160	Direct M-LASSO	0.0027 (0.0050)	0.1592	1.020 (0.007)	0.020 (0.007) [0.006, 0.033]
Arm-specific cancellation	160	Arm-specific LASSO (AIPW)	0.0018 (0.0052)	0.1647	1.091 (0.019)	0.091 (0.019) [0.054, 0.128]
Arm-specific cancellation	160	DGP oracle	0.0029 (0.0050)	0.1576	1.000 (0.000)	0.000 (0.000) [0.000, 0.000]
Arm-specific cancellation	240	DM	0.0091 (0.0040)	0.1263	1.000 (0.000)	0.000 (0.000) [0.000, 0.000]
Arm-specific cancellation	240	All-X OLS	0.0066 (0.0052)	0.1634	1.675 (0.065)	0.675 (0.065) [0.548, 0.801]
Arm-specific cancellation	240	All-X Lin	0.0069 (0.0051)	0.1602	1.611 (0.061)	0.611 (0.061) [0.490, 0.731]
Arm-specific cancellation	240	One-step LASSO	0.0096 (0.0040)	0.1276	1.022 (0.007)	0.022 (0.007) [0.009, 0.035]
Arm-specific cancellation	240	Direct M-LASSO	0.0093 (0.0040)	0.1269	1.011 (0.006)	0.011 (0.006) [-0.001, 0.022]
Arm-specific cancellation	240	Arm-specific LASSO (AIPW)	0.0084 (0.0041)	0.1288	1.040 (0.014)	0.040 (0.014) [0.013, 0.067]
Arm-specific cancellation	240	DGP oracle	0.0091 (0.0040)	0.1263	1.000 (0.000)	0.000 (0.000) [0.000, 0.000]
Arm-specific cancellation	400	DM	-0.0033 (0.0031)	0.0991	1.000 (0.000)	0.000 (0.000) [0.000, 0.000]
Arm-specific cancellation	400	All-X OLS	-0.0044 (0.0037)	0.1165	1.381 (0.044)	0.381 (0.044) [0.296, 0.467]
Arm-specific cancellation	400	All-X Lin	-0.0056 (0.0035)	0.1111	1.256 (0.033)	0.256 (0.033) [0.191, 0.321]
Arm-specific cancellation	400	One-step LASSO	-0.0035 (0.0031)	0.0996	1.009 (0.004)	0.009 (0.004) [0.001, 0.018]
Arm-specific cancellation	400	Direct M-LASSO	-0.0032 (0.0031)	0.0993	1.004 (0.004)	0.004 (0.004) [-0.003, 0.012]
Arm-specific cancellation	400	Arm-specific LASSO (AIPW)	-0.0039 (0.0032)	0.1011	1.041 (0.010)	0.041 (0.010) [0.020, 0.061]
Arm-specific cancellation	400	DGP oracle	-0.0033 (0.0031)	0.0991	1.000 (0.000)	0.000 (0.000) [0.000, 0.000]

Notes: Bias is calculated relative to the realized finite-population ATE. Relative MSE uses the difference in means as its within-cell benchmark. The oracle gap is in difference-in-means MSE units. Parentheses contain Monte Carlo standard errors; brackets in the final column contain untruncated 95% Monte Carlo intervals based on paired replication-level influence values. Each newly simulated cell contains 1,000 replications; the $N = 400$ entries reuse the baseline replications.